Physics of Living Systems

Design principles of transcription factors with intrinsically disordered regions

Reviewed Preprint
v2 • August 27, 2025
Revised by authors

Reviewed Preprint
v1 • April 14, 2025

Wencheng Ji, Ori Hachmo, Naama Barkai, Ariel Amir

Department of Physics of Complex Systems, Weizmann Institute of Science, Rehovot, Israel • John A. Paulson School of Engineering and Applied Sciences, Harvard University, Cambridge, United States • Department of Molecular Genetics, Weizmann Institute of Science, Rehovot, Israel

eLife Assessment

This paper presents an **important** theoretical exploration of how a flexible protein domain with multiple DNA binding sites may simultaneously provide stability to the DNA-bound state and enables exploration of the DNA strand. The authors propose a mechanism ("octopusing") for protein doing a random walk while bound to DNA which simultaneously enables exploration of the DNA strand and enhances the stability of the bound state. This study presents **compelling** evidence that their findings has implications for the way intrinsically disordered regions (IDR) of transcription factors proteins (TF) can enhance their ability to efficiently find their binding site on the DNA from which they exert control over the transcription of their target gene. The paper concludes with a comparison of model predictions with experimental data which gives further support to the proposed model.

https://doi.org/10.7554/eLife.104956.2.sa3

Abstract

Transcription Factors (TFs) are proteins crucial for regulating gene expression. Effective regulation requires the TFs to rapidly bind to their correct target, enabling the cell to respond efficiently to stimuli such as nutrient availability or the presence of toxins. However, the search process is hindered by slow diffusive movement and the presence of 'false' targets - DNA segments that are similar to the true target. In eukaryotic cells, most TFs contain an Intrinsically Disordered Region (IDR), which is commonly assumed to behave as a long, flexible polymeric tail composed of hundreds of amino acids. Recent experimental findings indicate that the IDR of certain TFs plays a pivotal role in the search process. However, the principles underlying the IDR's role remain unclear. Here, we reveal key design principles of the IDR related to TF binding affinity and search time. Our results demonstrate that the IDR significantly enhances both of these aspects. Furthermore, our model shows good agreement with experimental results, and we propose further experiments to validate the model's predictions.

Introduction

Transcription Factors (TFs) are fundamental proteins that regulate gene expression: they bind to specific DNA sequences to control gene transcription [1]; they respond adaptively to various cellular signals, thereby modulating gene expression [2, 3]; they guide processes such as cell differentiation and development, and their malfunctions are often linked to disease onset [4, 5]. The principles of gene regulation were first formulated in the 1960s through pioneering studies of the *E. coli* Lac operon [6], establishing that the regulation mechanism is via the binding of a TF to the regulatory DNA regions associated with the genes, the TF's target. Effective regulation requires TFs to have sufficient binding affinity and a binding rate comparable to cellular processes. Despite these decades of research, the factors determining the binding strength and search efficiency of eukaryotic TFs remain incompletely understood [7, 8]. In eukaryotic cells, besides having a DNA-binding domain (DBD), approximately 80% of TFs also have one or more intrinsically disordered regions (IDRs), which lack a stable 3D structure and can be assumed as flexible, unstructured polymers, hundreds of amino acids (AAs) in length [9–12]. Eukaryotic DBDs often exhibit weaker affinity to their recognition sequences compared to their bacterial counterparts, indicating that IDRs are necessary for achieving stable binding [13]. However, IDRs are challenging to study because their functions can be maintained despite rapid sequence divergence, complicating traditional sequence-based analyses [14, 15]. Molecular dynamics simulations suggest that IDRs play a significant role in promoting target recognition [16], by increasing the area of effective interaction between the TF and the DNA. Upon binding, the search problem becomes effectively a one-dimensional diffusion process [17]. Recent *in vivo* experiments done in yeast also suggest that IDRs could take a key part in guiding TFs to the DNA regions containing their targets [18, 19]. Truncation studies of IDRs in various TFs have demonstrated that removing these regions often results in reduced binding affinity, suggesting they contribute significantly to stabilizing TF-DNA interactions [18]. Different mechanisms have been proposed in the literature to explain IDR contributions: some suggest non-specific electrostatic interactions [20], others propose phase separation that concentrates TFs near their targets [21], while a third view suggests specific IDR-DNA interactions analogous to those of structured domains [22].

The problem of TF search has been widely studied, theoretically, albeit with the majority of works focusing on parameter regimes relevant for bacteria. One important mechanism is that of facilitated diffusion, in which the TF performs 1D diffusion along the DNA and at any given moment can fall off and perform 3D diffusion until reattaching to the DNA (a 3D excursion). The 1D diffusion along the DNA and the 3D excursions alternate until the TF finds its target. This mechanism is well characterized theoretically [23–29] and supported by experiments in *E. coli* [30]. However, due to the significant structural differences, including larger genome size and chromatin structure, the facilitated diffusion mechanism cannot explain the fast search times observed in eukaryotes, and a qualitatively different mechanism is required [7, 19, 31–33].

Here, we attempt to address the fundamental question of how IDRs can enhance the binding affinity of the TFs and speed up their search process, and to discover the associated design principles in eukaryotes. The problem of binding affinity (binding probability) exhibits the classical trade-off between energy and entropy: on one hand, the binding of IDR to the DNA is energetically favorable. On the other hand, this binding leads to constraints on the positions of the IDR (pinned to the DNA at the points of attachment) which leads to a reduction in the entropy associated with the possible polymer conformations - hence an *increase* in the free energy of the system. Therefore the effects of the disordered tail on affinity are non-trivial, leading to physical problems reminiscent of those associated with the well-studied problem of DNA unzipping [34]. We find that the binding probability increases dramatically for TFs having IDRs for a broad parameter regime. The optimal search process consists of one round of 3D-1D search, where at first the TF performs 3D diffusion and then binds the antenna and performs 1D diffusion until finding its target, rarely unbinding for another 3D excursion. This process is presented in Fig. **1(a, c)**. We obtain an expression for the search time that shows good agreement with our numerical results.

Results

Binding probability of the TF from the equilibrium approach

The experiment described in Ref. [18] studied the binding affinity of wild-type and mutated TFs to their corresponding targets. They found that shortening the IDR length results in weaker affinity. To qualitatively model the binding affinity of a TF to its target, we utilize a well-known physical model of polymers, the Rouse Model [35, 36] (used in the subsequent dynamical section as well). See Materials and Methods. We note that our conclusions do not hinge on this choice, and our results hold also when using other polymer models (see Supplementary S1). Since the IDR is thought to recognize a relatively short flanking DNA sequence of the DBD target [18], we model a DNA segment as a straight-line "antenna" region [26, 37–39], where the IDR targets reside. We assume that the interaction between TF and DNA outside the antenna region is negligible. Note that the interaction between IDR and DNA is a specific interaction, with genomic localization determined by multivalent interactions mediated by hydrophobic residues [40, 41].

To gain an intuition into the binding probability of TF, we first study a TF with a single IDR site. From here on, energies will be measured in units of $k_B T$, to simplify our notation. We assume that the IDR has only one corresponding target. There are four possible states: neither the DBD nor the IDR are bound, the IDR is bound, the DBD is bound, and both are bound, which correspond to P_{free}, PIDR, PDBD , and Pboth, respectively.

The probability that the TF is bound at the DBD target is the sum of P_{DBD} and P_{both}. In thermal equilibrium, each of the 4 terms is given by the Boltzmann distribution, which can be solved analytically by integrating-out the additional degrees of freedom associated with the polymer (Supplementary S2). To better understand the associated design principles, it will be helpful to compare this probability with that of a "simple" TF (i.e., without the IDR), which we denote by P_{simple}. We find that:

$$\frac{P_{\mathrm{TF}}}{P_{\mathrm{simple}}} \approx (1+\mathcal{P}), \tag{1}$$

where $\boldsymbol{P}$ is an enhancement factor, governing the degree to which the IDR improved the binding affinity. Importantly, we find that:

$$\mathcal{P}(E_B, d, l_0) = e^{E_B} V_1 f(d, l_0), \tag{2}$$

where $f(d,l_0) = \left(\frac{3}{2\pi l_0^2}\right)^{3/2} \exp\left(-\frac{3d^2}{2l_0^2}\right)$, is the probability density, with units of inverse volume, V_1 is the targets' volume, on the order of 1-10bp^3, as shown in Fig. **1c**, -E_B is the binding energy for per IDR target (E_B is always positive), d is the distance between the DBD target and the IDR target, and l_0 is the distance between the DBD and the IDR binding site on the TF (l_0 and d are also the distances for neighboring IDR targets and neighboring IDR sites, respectively, in the case of multiple IDR targets and sites). This shows that the presence of the IDR always increases the

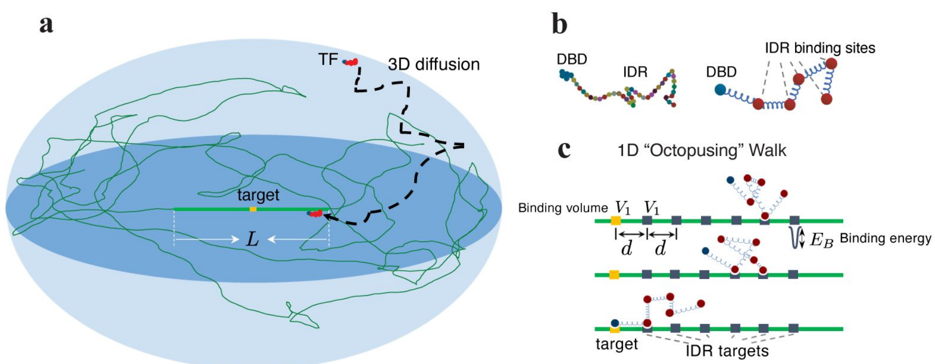

Figure 1

Model illustration of a TF with the IDR and its search process.

a, Illustration of a TF locating its target site (highlighted in orange) on the DNA strand (depicted as a green curve). Surrounding the target is a region of length L, where the TF's IDR interacts with the DNA. The TF performs 3D diffusion until it encounters and binds to this "antenna" region. **b**, Illustration of a TF that includes an IDR composed of AAs. On the right: a model of the TF, a polymer chain, comprising of a single binding site on the DNA Binding Domain (DBD), and multiple binding sites on the IDR (also known as short linear motifs [8]). **c**, Once bound to the antenna, the TF performs effective 1D diffusion until it reaches its target. The 1D diffusion is via the binding and unbinding of sites along the IDR, a process we coin "octopusing". V_1 is the targets' volume, d the separation, and E_B the binding energy per binding site, where each site corresponds to a short linear motif.

binding affinity since $\mathscr{P}$ is always positive. However, the magnitude of enhancement can be negligible: For instance, even if E_B is large, the term $f(d, l_0)$, associated with the entropic penalty (because the end-to-end distance is fixed, reducing the degrees of freedom compared to a free polymer), could lead to small values of $\mathscr{P}$. The exponential decay of $\mathscr{P}$ with d^2 (through the term $f(d, l_0)$), indicates that placing the IDR targets close to the DBD target is preferable. However, due to physical limitations, d cannot be arbitrarily small and in reality will have a lower bound. Once d is fixed, $\mathscr{P}$ is maximal at $l_0 = d$, as can be gleaned from equation **(2)**. This condition minimizes the free energy penalty associated with the entropic contribution of the term $f(d, l_0)$. As we shall see, this design principle is also intact when multiple IDR sites and targets are involved, which is the scenario we discuss next.

A *priori*, it is unclear whether having more binding sites ($n_b > 1$) on the IDR or more targets ($n_t > 1$) on the DNA will be beneficial or not. We therefore sought to investigate how the binding affinity depends on these parameters, as well as the binding energy and positions of the targets and binding sites. This generalization requires taking into account all the possible number of bindings, n (ranging from 0 to min (n_b, n_t)), and all possibilities for binding orientations for each n, $\binom{n_b}{n}\binom{n_t}{n}n!$ distinct configurations in total. For each of them, we may analytically integrate-out the polymer degrees of freedom and calculate explicitly the contribution to the partition function (Supplementary S2). Summing these terms together, numerically, allows us to explore how the different parameters affect the binding specificity, which we describe next. To characterize the binding affinity, we resort again to the ratio $Q = P_{\mathrm{TF}}/P_{\mathrm{simple}}$, measuring the effectiveness of the binding in comparison to the case with no IDRs. The length of TFs in our considerations, which varies with n_b, is comparable to that in experiments [18].

We find the design principle: $l_0 = d$, identical to what we showed for the case of a single IDR binding site. Specifically, Q is maximal at $l_0 \sim d$ for all combinations of n_b and n_t. Fig. **2a** displays this principle for several combinations of n_b and n_t. All combinations are provided in **Supplementary Fig. S3**. This is plausible since for the IDR to be bound to the DNA at multiple places without paying a large entropic penalty, the distance between target sites should be compatible with the distance between binding sites. We therefore set $l_0 = d$ and search for other design principles. Fig. **2b** shows a heatmap of Q varying with n_b and n_t. At large n_t, as n_b increases, the value of P_{TF} exhibits an initial substantial increase, followed by a gradual rise until it reaches saturation (shown explicitly in **Supplementary Fig. S4**). A similar behavior is also observed when varying the value of n_t. (When n_t exceeds n_b, the contribution from the states where the IDR is bound but the DBD is unbound increases with increasing n_t, which slightly lowers P_{TF}; see **Supplementary Fig. S4**.) In fact, the heatmap shown in Fig. **2b** shows that the value of P_{TF} is governed, approximately, by the *minimum* of n_b and n_t (this is reflected in the nearly symmetrical “L-shapes” of the contours). This places strong constraints on the optimization, and suggests that there will be little advantage in increasing the value of either n_b or n_t beyond the point $n_b = n_t$, revealing another design principle (under the assumption that it is preferable to have smaller values of n_b and n_t for a given level of affinity, in order to minimize the cellular costs involved). We verify this principle holds at additional values of E_B. Fig. **2c** shows that the contours corresponding to $P_{TF} = 0.9$ at varying values of E_B all take an approximate symmetrical L-shape, thus supporting the design principle $n_b = n_t$. For a given E_B, we denote the optimal values as $n_b^* = n_t^*$, and show them as hexagrams in the plot. We further find that n_b^* decreases with increasing E_B, as displayed in the inset. These relationships also hold at $P_{TF} = 0.5$ (Supplementary S5).

We examine the variation of Q with respect to E_B while keeping n_b = n_t constant. In Fig. **2d**, as the binding energy E_B increases, Q increases from a value of ~ 1 to its saturation value, $1/P_{\text{simple}}$, in a switch-like manner: i.e., there is a characteristic energy scale, E_{th}. To estimate E_{th}, when $Q \ll 1/P_{\text{simple}}$ (i.e., P_{TF} 1), we approximate the expression of Q as:

$$Q \approx 1 + \Sigma_{n=1}^{n_b} \Sigma_{\text{orn}} \prod_{i=1}^{n} \mathcal{P}(E_B, r_{\text{orn},i}, l_{\text{orn},i}), \tag{3}$$

where $l_{\text{orn},\,i}$ and $r_{\text{orn},\,i}$ denote the segment length between two adjacent bound sites and the distance associated with their respective targets, respectively, and "orn" represents the orientation of the given configuration (Supplementary S2). **Equation (3)** reduces to **equation (1)** when n_t = n_b = 1. To substantially increase Q, the configuration that contributes the most to Q, would satisfy $(\mathscr{H}(E_{\text{B}}, d, d))^{n_b} \ll 1$. Thus, the characteristic energy E_{th} could be estimated using the formula $\mathscr{H}(E_{\text{th}}, d, d) = 1$, resulting in:

$$E_{\text{th}} = \frac{3}{2}\left[\ln(\frac{2\pi}{3\phi^2}) + 1\right], \tag{4}$$

where $\phi = V_1^{1/3}/d$ represents the proportion of IDR targets within the antenna. **Equation (4)** is a direct consequence of the energy-entropy trade-off. Increasing Φ leads to a decrease in entropy loss due to binding the IDR, subsequently reducing the demands on binding energy. Given that Φ^2 = 0.02, E_{th} is about $8.5k_BT$. The vertical dashed line in Fig. **2d**, obtained from **equation (4)**, effectively captures the region of significant increase for large values of n_b. Hence, we obtain the third design principle: $E_B > E_{\text{th}}$, which ensures the significant binding advantage of TFs with IDRs.

So far, we have found that the binding design principles for TFs with IDRs are as follows: (**i**) target distance d should be as small as possible, (**ii**) IDR segment binding distance l_0 should be comparable to d, (**iii**) it is preferable to have fewer binding sites n_b^* and binding targets n_t^* as the binding strength E_B increases, with $n_b^* \sim n_t^*$, and (**iv**) E_B should be larger than a threshold that solely depends on the proportion Φ of IDR targets within the antenna, as indicated by **Eq. (4)**. In our estimation, with Φ^2 = 0.02, we obtain $E_{\text{th}} \approx 8.5k_BT$. The above principles have also been validated in scenarios involving Poisson statistics of binding site distances and target distances, thus indicating their robust applicability (**Supplementary Fig. S6**).

Search time of the TF from the dynamics

In this section, we study the design principles of the TF *search time*, a dynamic process. Here, we confine the TF with $\tilde{n} = n_b+1$ sites (n_b IDR sites and one DBD site) to a sphere of radius R, the cell nucleus, and use over-damped coarse-grained molecular dynamics (MD) simulations. The sites and targets interact via a short-range interaction. The DBD target with a radius a is placed at the sphere's center. See the sketch in Fig. **1a**. Once the DBD finds its target, the search ends. We denote the mean total search time (i.e., the mean first passage time, MFPT), as t_{total}. We estimate it by averaging 500 simulations, each starting from an equilibrium configuration of the TF at a random position in the nucleus. Simulation details are provided in the Materials and Methods section.

By conducting comprehensive simulations of t_{total} across various parameter values of L, E_B, and $\tilde{n}$, we were able to identify its minimum value, which is approximately at L^* = 300bp, E_B^* = 11, and $\tilde{n}^*$ = 4 for R = 500bp ≈ 1/6μm. Here, one base pair (bp) is equal to 0.34nm. To speed up the simulations, we utilized several computational tricks, including adaptive time steps and treating the TF as a point particle when it is far from the antenna (see Materials and Methods). Despite of this, the simulations require 10^5 CPU hours due to the IDR trapped in the binding targets on the

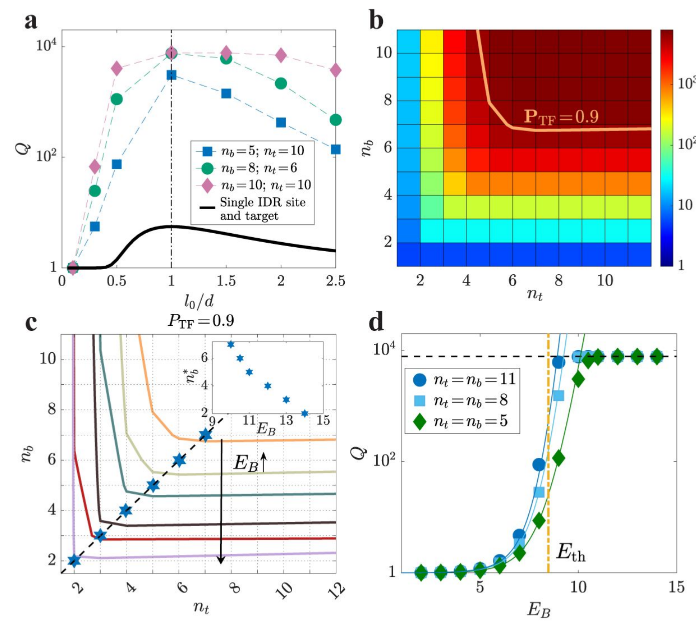

Figure 2

Design principles across various parameters: l_0/d, n_b, n_t, and E_B.

a, Plots of $Q = P_{TF}/P_{simple}$ varying with l_0/d are shown for several combinations of n_t and n_b for typical values of the relevant quantities: the nucleus volume is $1\mu m^3$, $E_{\mathrm{DBD}} = 15$, $E_B = 10$, $V_1 = (0.34\mathrm{nm})^3$, and $d = \sqrt{50} \cdot 0.34\mathrm{nm}$. **b**, A heatmap of Q vs. n_b and n_t at $E_B = 10$ and $d = l_0$. **c**, Contour lines at different E_B at $P_{\mathrm{TF}} = 0.9$. Blue hexagrams represent the design principle $n_b^* = n_t^*$, with the black dashed line as a visual guide. The inset shows the dependence of n_b^* on energy. **d**, Q for varying E_B at $d = l_0$ for several $n_b = n_t$. The vertical line represents E_{th} as determined by **equation (4)**, the solid curves illustrate the approximation of Q obtained from **equation (3)**, and the horizontal line indicates where $P_{\mathrm{TF}} = 1$.

antenna, which results in the slow dynamics as found in the previous simulation works [16, 42]. The probability distribution function (pdf) of individual search times exponentially decays with a typical time equal to t_{total}, see inset of **Fig. 3**. This indicates that t_{total} provides a good characterization of the search time.

We find that t_{total} varies non-monotonically with L at fixed E_B^* and $\tilde{n}^*$ as shown in **Fig. 3**. Once a TF attaches to one of its targets, it will most likely diffuse along the antenna until reaching the DBD target (Supplementary S7). It means that the optimal search process is a single round of 3D diffusion followed by 1D diffusion along the antenna. In Supplementary S8, by analytically calculating the mean total search time using a coarse-grained model, we find that a single round of 3D-1D search is typically the case in the optimal search process. Note that our model differs profoundly from the conventional facilitated diffusion (FD) model in bacteria: In the latter, the TF is assumed to bind non-specifically to the DNA. Within our model, the antenna length L can be tuned by changing the placement of the IDR targets on the DNA, making it an adjustable parameter rather than the entire genome length. As opposed to the FD model, which requires numerous rounds of DNA binding-unbinding to achieve minimal search time, within our model, the minimal search time is achieved for a single round of 3D and 1D search.

Interestingly, when we observe the motion of a TF along the antenna, IDR sites behave like “tentacles”, as illustrated in Fig. **1c**, constantly binding and unbinding from the DNA. We refer to this 1D walk as “octopusing”, inspired by the famous “reptation” motion in entangled polymers suggested by de Gennes [43] (see Supplementary Video). We note that previous numerical studies [16, 17] used comprehensive, coarse-grained, MD simulations to demonstrate that including heterogeneously distributed binding sites on the IDR could facilitate intersegmental transfer between different regions of the DNA, which is reminiscent of our octopusing scenario.

The mean 3D search time, t_{3D}, and the mean 1D search time, t_{1D} are also shown in **Fig. 3**. To evaluate t_{3D}, we take advantage of several insights: first, if the TF gyration radius, $r_p = \sqrt{\tilde{n}/6}l_0$, is comparable or larger than the distance between targets, d, we may consider the chain of targets as, effectively, a continuous antenna. Making the targets denser would hardly accelerate the search time, reminiscent of the well-known chemoreception problem where high efficiency is achieved even with low receptor coverage [44]. Second, when the distance between the TF and the antenna is smaller than r_p, one of the binding sites on the TF would almost surely attach to the targets. This allows us to approximately map the problem to the MFPT of a particle hitting an extremely thin ellipsoid with long-axis of length L and radius r_p, the solution of which is $\ln(L/r_p)/L$ [45]. We find that this scaling captures well the dependence of t_{3D} on the parameters, with a prefactor $\alpha \sim O(1)$. For the 1D search from a single particle picture, since the TF is unlikely to detach once it attaches, it performs a 1D random walk. So, taken together, the mean total search time is

$$t_{\text{total}} = t_{\text{3D}} + t_{\text{1D}} \approx \alpha \frac{\tilde{n} R^3}{DL} \ln(L/r_p) + \beta \frac{L^2}{D}, \tag{5}$$

where D is the 3D diffusion coefficient of one site. For a TF with $\tilde{n}$ sites, its 3D diffusion coefficient, D_3, is $D/\tilde{n}$. α and β are fitting parameters. The semi-analytical formulas of t_{1D}, t_{3D}, and t_{total}, denoted by the solid lines in **Fig. 3**, agree well with the simulation results for $\alpha = 0.5$ and $\beta = 14.5$. The theoretical result for the time spent in 3D was originally calculated for a point searcher [45]. In that case, a value of $\alpha = 2/3$ was obtained. β is larger at larger E_B (see Supplementary S7).

We also examine the dependence of t_{total} on the key parameters *EB* and $\tilde{n}$, in the vicinity of the optimal parameters which minimize the total search time (Supplementary S7). We find that the affinity of TFs to the antenna decreases rapidly with decreasing the IDR length [18] and E_B. Consequently, when $\tilde{n} < \tilde{n}^*$ or $E_B > E_B^*$, multiple search rounds are needed to reach the DBD

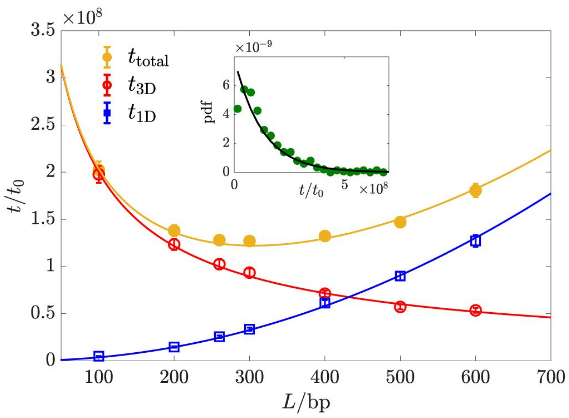

Figure 3

Mean search time varying with the antenna length *L*.

t_{total}, t_{3D} and t_{1D} vs. *L* at E_B^* and $\tilde{n}^*$. The solid curves correspond to t_{total}, t_{3D} and t_{1D} in equation (5). Inset: probability density function (pdf) exponentially decays at large *t*. pdf ≈ $\exp(-t/t_{\text{total}})/t_{\text{total}}$ (black solid line). The unit t_0 represents the time for one AA to diffuse 1bp. a = 0.5bp, *d* = 10bp, and l_0 = 5bp.

target, resulting in a significant escalation of t_{total}. When $\tilde{n}>\tilde{n}^*$, t_{total}, t_{3D}, and t_{1D} agree with equation (5). When $E_B > E_B^*$, since t_{3D} is independent of E_B, the variation in t_{total} stems from t_{1D} which exponentially increases with E_B.

The minimal search time for a simple TF is $t_{\text{simple}} = R^3/(3Da)$ [45]. For the TF with an IDR, by minimizing the expression in equation (5), we find that the optimal length approximately scales as $L^* \propto R$, and therefore the minimal search time, t_{min}, is approximately proportional to R^2. At R = 500 bp ≈ 1/6μm, the search time is more than ten times shorter than a simple TF, despite the latter's diffusion coefficient being several times larger. Based on our analytic result, we expect ~ 50 fold enhancement for the experimental parameters of yeast (note that we assume the scaling $t_{\text{min}} \sim R^2$ still holds in this regime; this formula is approximate since it does not take into account the persistence length of DNA [46], which is several-fold shorter than the optimal antenna length L^*. We have verified that reducing the DNA persistence length, which promotes increased DNA coiling, results in only a modest increase in mean search time. Even under extreme coiling conditions, the increase remains below 30% of the baseline value, as detailed in Supplementary S9. Converting tmin into a second-order on-rate yields approximately 10^8 — 10^9 M^{-1} s^{-1}, assuming $D \approx 10^{-13} m^2/s$ and a TF copy number of 100. We also verified that the 3D diffusion coefficient slightly changes under a complex DNA configuration, where a point-searcher TF non-specifically interacts with the entire DNA, indicating the robustness of our optimal search process (see Supplementary S10).

Next, we characterize the dynamics of the octopusing walk (Supplementary S11). During this process, different IDR binding sites constantly bind and unbind from the antenna, leading to an effective 1D diffusion. A *priori*, one may think that the TF faces conflicting demands: it has to rapidly diffuse along the DNA while also maintaining strong enough binding to prevent detaching from the DNA too soon. Indeed, within the context of TF search in bacteria, this is known as the speed-stability paradox [25]. Importantly, the simple picture of the octopusing walk explains an elegant mechanism to bypass this paradox, achieving a high diffusion coefficient with a low rate of detaching from the DNA. This comes about due to the compensatory nature of the dynamics: there is, in effect, a feedback mechanism whereby if only a few sites are bound to the DNA, the on-rate increases, and similarly, if too many are bound (thus prohibiting the 1D diffusion) the off-rate increases. Remarkably, this mechanism remains highly effective even when the typical number of binding sites is as low as two. Further details are provided in Supplementary S11.

Comparing with experimental results

In Ref. [18], the binding probability of the TF Msn2 in yeast was measured, where the native IDR was shortened by truncating an increasingly large portion of it. The results indicate that binding probability decreases with increasing truncation of the IDR, as illustrated in **Fig. 4(a)**. We utilized our model to calculate the relative binding probability compared to P_{TF} at a zero truncating length. Using the DBD binding energy *EDBD* as our only fitting parameter, our model achieves quantitative agreement with the experimental data. In **Fig. 4(b)**, we estimate the search time as a function of L for $R = 1\mu m$ and $D_0 = 10^{-10} m^2/s$ for a single AA. The estimated timescales are comparable to empirical results, which are 1 to 10 seconds [19, 47], as indicated by the shaded area. We have also estimated the on-rate k_{on} and off-rate k_{off} of the TF as they vary with IDR length, as shown in **Fig. 4(c)**. In the kinetic proofreading mechanism [48], specificity is associated with the off-rate, but recent studies [49] suggest that in other scenarios, specificity is encoded in the on-rate. Therefore, it is important to explore how specificity is encoded within this mechanism. Our results indicate that the variation in the dissociation constant $k_{\text{off}}/k_{\text{on}}$ is dominated by k_{off}, suggesting that specificity is encoded in the off-rate. The magnitudes and variations of these rates could be directly compared to experimental results in future studies.

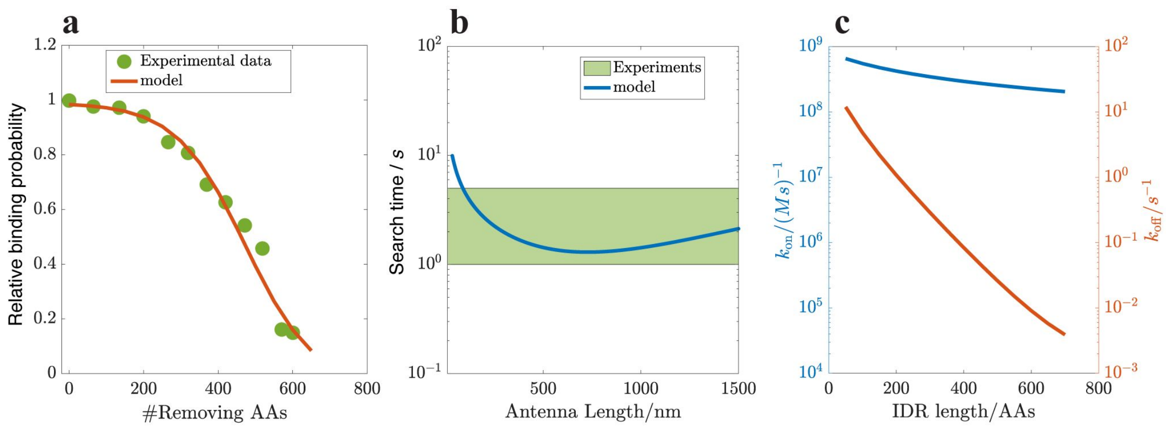

Figure 4

a, Relative binding probability varies with the truncation length of the IDR, which quantitatively agrees with the experimental data in Ref. [18], where we only vary E_{DBD} (binding energy of DBD) to ensure that the maximal value reaches 1 at zero truncation length. Antenna length L = 1000nm, E_{DBD} =22 and E_B = 11. Other parameters are set the same as in **Fig. 2**. **b**, The search time estimated by our model (from **Eq. (5)** and divided by a TF copy number of 100) is quantitatively comparable with the experimental results [47], as guided by the shaded area. **c**, The on-rate $kon = t_{\mathrm{total}}/V_c$ and off-rate $koff = (1 - P_{\mathrm{TF}})/(P_{\mathrm{TF}} t_{\mathrm{total}})$ varying with the IDR length, indicating that $k_D = k_{\mathrm{off}}/k_{\mathrm{on}}$ ranges from 0.01 to 10 nM, with its variation primarily dominated by the off-rate.

Conclusion and discussion

In summary, we addressed the design principles of TFs containing IDRs, with respect to binding affinity and search time. We find that both characteristics are significantly enhanced compared to simple TFs lacking IDRs.

The simplified model we presented provides general design principles for the search process and testable predictions. One possible experiment that could shed light on one of our key assumptions is "scrambling" the DNA (randomizing the order of nucleotides). Our results depend greatly on our assumption that there exists a region on the DNA, close to the DBD target, that contains targets for the IDR binding sites. It is possible to shuffle, or even remove, regions of different lengths around the DBD target and to measure the resulting changes in binding affinity. According to our results, we would expect that shuffling a region smaller in length than the IDR should have minimal effect on the binding affinity and search time, but that shuffling a region longer than the IDR should have a noticeable effect.

Another useful test of our model would be to probe the *dynamics* of the TF search process by varying the waiting time before TF-DNA interactions are measured: Within the ChEC-seq technique [50] employed in Ref. [18], an enzyme capable of cutting the DNA is attached to the TF, and activated at will (using an external calcium signal). Sequencing then informs us of the places on the DNA at which the TF was bound at the point of activation. Measuring the binding probabilities at various waiting times — defined as the time between adding calcium ions and adding transcription factors, ranging from seconds to an hour, would inform us both of the TF search dynamics as well as the equilibrium properties: Shorter time scales would capture the dynamics of the search process, while longer time scales would reveal the equilibrium properties of the binding. At short waiting times (within several seconds), we would expect the binding probability to increase with waiting times. By analyzing a normalized binding probability curve as a function of the waiting time t, we could fit the curve using the function 1-exp($-t/t_{\text{search}}$). From this curve, we can infer the search time. Our results suggest that the presence of the IDR in a TF leads to increased binding probabilities over various waiting times and a reduced waiting time to reach a higher saturation binding probability. Furthermore, as shown in **Fig. 4** obtained from our model, k_{on} can be determined from the measurements of t_{search}. Using the binding probability, which is associated with $k_D = k_{\text{off}}/k_{\text{on}}$, k_{off} can also be calculated.

While our work primarily focused on coarse-grained modeling of the search process and the hypothesized octopusing dynamics, higher-resolution molecular dynamics simulations could provide more detailed molecular insights and further validate our proposed design principles [16, 17]. Since these principles are grounded in fundamental physical concepts, such as the energy-entropy trade-off, they are likely to be robust across diverse molecular systems. We anticipate their applicability to other biophysical contexts, such as nonspecific RNA polymerase binding [51] and multivalent antibody binding [52], where the associated dynamic processes would also be of significant interest for future research. Beyond our octopusing mechanism, other mechanisms may also contribute to the search and binding process. IDRs could facilitate complex formation and cooperativity between multiple transcription factors, as suggested by studies showing co-localization at common target promoters [53]. IDRs might also enhance efficiency by directing transcription factors to specific nuclear compartments or biomolecular condensates, thereby reducing the effective search space and time [21, 54, 55]. These complementary mechanisms [7, 8, 56] suggest that our findings represent one component of a broader regulatory framework governing transcriptional control in eukaryotic cells.

Materials and methods

Model

The Rouse Model describes a polymer as a chain of point masses connected by springs and characterized by a statistical segment length l_0 resulting from the rigidity of springs k and the thermal noise level k_BT, $l_0^2 = 3k_BT/k$ [35, 36]. We coarse-grain the TF by representing it with a few binding sites for the IDR, and an additional site for the DBD, as illustrated in **Fig. 1b**. The targets are assumed to be positioned at equal spacing, with one target for the DBD and a few targets for the IDR, corresponding to the orange and black squares in **Fig. 1c**. We have also verified that relaxing this assumption, and using randomly positioned targets, does not affect our results (Supplementary S6). In the Rouse Model, the 3D probability density describing the relative distance in the equilibrium state, denoted as r, between any two sites can be derived through Gaussian integration [36], leading to:

$$f(r,l) = \left(\frac{3}{2\pi l^2}\right)^{\frac{3}{2}} \exp\left(-\frac{3r^2}{2l^2}\right), \tag{6}$$

where $l = \sqrt{n_{\mathrm{seg}}}\, l_0$ and n_{seg} is the number of segments between the two sites.

Dynamical simulations

Each binding site of the TF at position r_i follows over-damped dynamics, which reads:

$$\Delta r_i = \frac{D}{k_BT} f(\{r_i\})\Delta t + \sqrt{2D\Delta t}\,\xi, \tag{7}$$

where the random variable ξ follows the standard normal distribution. D is the diffusion coefficient of the coarsegrained binding site. The force acting on each site is $\boldsymbol{f}(\{\boldsymbol{r}_i\}) = -\partial U_{total}/\partial r_i$, and

$$U_{\mathrm{total}} = \sum_{i=0}^{n_b-1} \frac{1}{2} k (r_{i+1} - r_i)^2 + \sum_{i,j} E_B \exp\left[-\frac{(r_i - R_j)^2}{2w_d^2}\right], \tag{8}$$

where R_j are the positions of the targets. w_d = 1bp is the typical width of the short-range interaction between sites and their corresponding targets. The typical distance l0 between coarse-grained binding sites is 5bp. The stiffness of the springs is thus $3k_BT/l_0^2 = 3k_BT/25\mathrm{bp}^2$. To speed up the simulations, when the sites' minimal distance to the antenna is larger than the threshold 10bp, the TF is regarded as a rigid body, and all sites move in sync with their center of mass. We verified that the resulting search time is insensitive to the value of this threshold. When the sites' minimal distance to the antenna is less than 10bp, we consider the excluded volume effect to ensure that different binding sites are not trapped at the same target. Specifically, we use the WCA interaction [57]:

$$U_{\mathrm{WCA}} = \begin{cases} \frac{\sigma^{12}}{|r_i - r_j|^{12}} - \frac{2\sigma^6}{|r_i - r_j|^6} + 1, & |r_i - r_j| < \sigma \\ 0, & |r_i - r_j| \geq \sigma \end{cases} \tag{9}$$

where σ = 1bp. In this region, we also adopted an adaptive time step: $\Delta t/(\mathrm{bp}^2/D)$ = $\min\left\{1, \frac{k_B T}{\max|f_i|\,\mathrm{bp}}\right\}\times 0.01$. Note that this choice ensures that the displacement of the sites at every step is much smaller than the binding site size w_d. Due to the IDR trapped in the binding targets on the antenna, which results in slow dynamics, the simulation time becomes significantly longer, as found in previous simulation studies [16, 42]. This slowdown is similar to the behavior observed in glass-forming liquids [58]. As a result, the individual simulation time steps range from 10^9 to 10^{11}.

Declarations

- The authors declare no competing interests.
- Data and code availability: The data supporting this study's findings and the codes used in this study are available from W. Ji and the corresponding author.
- Authors' contributions: All authors contributed to the design of the work, analysis of the results, and the writing of the paper.

Acknowledgements

We thank Samuel Safran, David Mukamel, Yariv Kafri, Yaakov Levy, Luyi Qiu, and Zily Burstein for insightful discussions. We thank the Clore Center for Biological Physics for their generous support.

Supplementary Materials: Design principles of transcription factors with intrinsically disordered regions

S1. Broad applicability of probability density *f (r, l)*

We verify the broad applicability of Eq. [1] in the main text, regardless of the specific details of microscopic interactions. To demonstrate this, we only require that the density function of the end-to-end distance *r* for a segment containing *m* amino acids (AAs), denoted as $f(r, l_0)$, complies with Eq. [1]. Specifically, it is.

$$f(r, l_0) \approx \left(\frac{3}{2\pi l_0^2}\right)^{3/2} \exp\left(-\frac{3r^2}{2l_0^2}\right), \tag{S1}$$

where l_0 is the typical length of the segment. When *m* is large, according to the central limit theorem, *r* should conform to the distribution in Eq. [S1] with a variance of $l_0^2 = m a_{\mathrm{eff}}^2$, where a_{eff} represents the effective length of an individual AA.

Next, we will consider a specific form of interaction, of springs with a finite rest-length. In one limit (where the rest length vanishes), this corresponds to the Rouse model (where Eq. [S1] is exact for any value of *m*, not only asymptotically) and in another limit (where the springs are very stiff) this corresponds to a chain of rods. As we shall see, Eq. [S1] provides an excellent approximation already for moderate values of *m*.

To proceed, we consider two adjacent AAs at positions $\vec{r}_i$ and $\vec{r}_{i+1}$ with an interaction denoted by $u(|\vec{r}_i - \vec{r}_{i+1}|)$. Their probability density is $\rho(\vec{r}_i, \vec{r}_{i+1}) \propto \exp\left(-u(|\vec{r}_i - \vec{r}_{i+1}|)/(k_B T)\right)$. $a_{\rm eff}^2$ can be calculated using $\rho(\vec{r}_i, \vec{r}_{i+1})$:

$$a_{\rm eff}^2 = \int d\vec{r} \int d\vec{r}_i \int d\vec{r}_{i+1} \, |\vec{r}_i - \vec{r}_{i+1}|^2 \rho(\vec{r}_i, \vec{r}_{i+1}) \delta\left(\vec{r} - (\vec{r}_i - \vec{r}_{i+1})\right). \tag{S2}$$

Specifically, for a spring-like interaction with a stiffness constant k_0 and a rest length b_0, the interaction is given by $u = \frac{1}{2} k_0 \left(|\vec{r}_i - \vec{r}_{i+1}| - b_0\right)^2$. This formula can also be interpreted as a harmonic approximation of the interaction around the equilibrium distance b_0. In this case, $a_{\rm eff}^2$ is derived analytically:

$$a_{\rm eff}^2 = a_0^2 + b_0^2 + \frac{2 b_0^2 a_0^2}{a_0^2 + 3 b_0^2}\left(1 + \frac{2 a_0^3}{6 b_0^2 a_0 + \sqrt{6\pi} b_0 \left(a_0^2 + 3 b_0^2\right)\left(1 + \mathrm{Erf}\left(\sqrt{\frac{3}{2}} \frac{b_0}{a_0}\right)\right) \exp\left(\frac{3 b_0^2}{2 a_0^2}\right)}\right), \tag{S3}$$

where $a_0^2 \equiv 3 k_0 / k_B T$ and $\mathrm{Erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z \exp(-x^2/2) dx$ is the error function. When $b_0 = 0$, we have $a_{\rm eff} = a_0$, and Eq. [S1] holds for any m. When $b_0 \gg a_0$, then $a_{\rm eff} \approx b_0$ corresponds to a chain of rods where the angle between consecutive rods is free to rotate. **Fig. S1(a)** shows that the numerical results obtained from this harmonic interaction are in good agreement with **Eq. [S1]** even at moderate values of m. The inset displays the variation of $a_{\rm eff}^2 / a_0^2$ with b_0/a_0 obtained from **Eq. [S3]**. In **Fig. S1(b)**, with $b_0 = a_0$, $f(r, l_0)$ rapidly converges to **Eq. [S1]**.

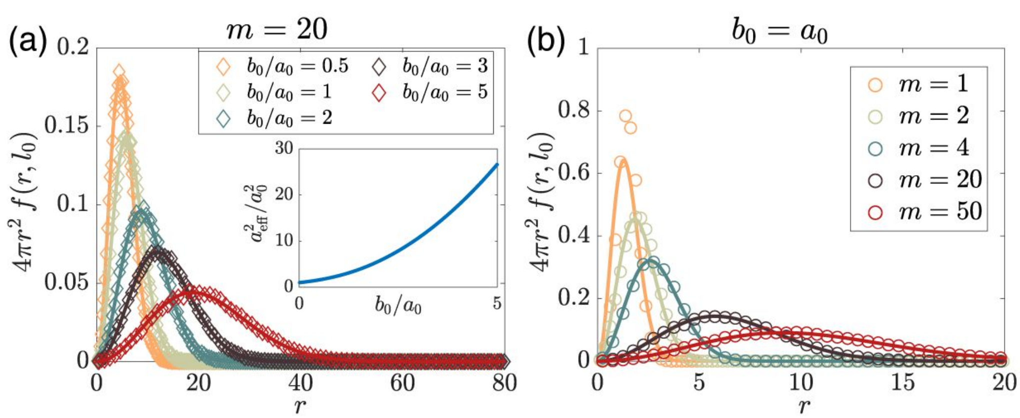

Figure S1

***f*(*r*, l_0) aligns well with Eq. [S1] (solid curves) for moderate *m* at various b_0.**

Inset: $a_{\rm eff}^2 / a_0^2$ vs. b_0/a_0 from **Eq. [S3]**. (b) **Eq. [S1]** serves as a good approximation (solid lines) even at small *m*.

S2. Derivation of the binding probability

In this section, we derive the generalized binding probability P_{TF} for n_b IDR binding sites and n_t IDR targets.

To achieve this generalization, we consider all the possible number of bindings, n from 0 to $\min(n_b, n_t)$, and all possibilities for binding orientations for each n. Note that the energy and entropy trade-off is naturally included in our model, and the generalized formula of P_{TF} is also applicable for non-uniformly distributed binding targets and binding sites.

Based on the Rouse model, the system's energy can be expressed as $\frac{k}{2}\sum_{i=1}^{n_b}(\vec{r}_i-\vec{r}_{i-1})^2 = \frac{3k_BT}{2l_0^2}\sum_{i=1}^{n_b}(\vec{r}_i-\vec{r}_{i-1})^2$, where $\vec{r}_i$ represents the position of the binding site i. The probability density in the absence of binding reads:

$$\rho(\vec{r}_0,\dots,\vec{r}_{n_b}) = \frac{1}{V_c}\left(\frac{3}{2\pi l_0^2}\right)^{\frac{3}{2}n_b} e^{-\frac{3}{2l_0^2}\sum_{i=1}^{n_b}(\vec{r}_i-\vec{r}_{i-1})^2}, \tag{S4}$$

where $\int d^3\vec{r}_0 \dots \int d^3\vec{r}_{n_b}\rho(\vec{r}_0,\dots,\vec{r}_{n_b}) = 1$.

Now we consider a given configuration with n IDR sites bound, where IDR sites $q_1 < \dots < q_n$ are bound to targets $s_1, \dots, s_n$. The positions of these bound targets are at $\vec{R}_{s_1},\dots,\vec{R}_{s_n}$. The DBD site is denoted by $q_0 = 0$ and its target is denoted by $s_0 = 0$. There are $n_b - n$ unbound IDR sites, indicated by $\bar{q}_1 < \dots < \bar{q}_{n_b-n}$. The corresponding binding probability is given by:

$$P_{\text{bound}}(\text{orn}, n) = \frac{1}{Z}\prod_{i=0}^{n}\int_{\vec{r}_{q_i}\in V_1} d^3\vec{r}_{q_i}\prod_{\ell=1}^{n_b-n}\int_{\vec{r}_{\bar{q}_\ell}\notin V_1} d^3\vec{r}_{\bar{q}_\ell}\,\rho(\vec{r}_0,\dots,\vec{r}_{n_b})\, e^{E_{\text{DBD}}+nE_B}, \tag{S5}$$

where "orn" stands for the orientation of the given configuration, Z is the normalization factor, and E_{DBD} is the DBD site binding energy for its target. There are $\binom{n_b}{n}\binom{n_t}{n}n!$ possible binding orientations at n. The key insight in evaluating this integral is to realize that we can use the result of **Eq. [1]** of the main text to integrate out the degrees of freedom associated with the sites between q_{i-1} and q_i:

$$\prod_{\bar{q}_\ell=q_{i-1}+1}^{q_i-1}\left[\int d^3\vec{r}_{\bar{q}_\ell}\left(\frac{3}{2\pi l_0^2}\right)^{3/2}\exp\left(-\frac{3(\vec{r}_{\bar{q}_\ell}-\vec{r}_{\bar{q}_\ell-1})^2}{2l_0^2}\right)\right] = f\left(\left|\vec{r}_{q_i}-\vec{r}_{q_{i-1}}\right|, \sqrt{q_i-q_{i-1}}\,l_0\right). \tag{S6}$$

Since $r_{q_{i-1}}$ and r_{q_i} are confined to a very small volume V_1, to an excellent approximation we may simply evaluate $f\left(\left|\vec{r}_{q_i}-\vec{r}_{q_{i-1}}\right|, \sqrt{q_i-q_{i-1}}\,l_0\right)$ in **Eq. [S6]** at $\vec{r}_{q_{i-1}} = \vec{R}_{s_{i-1}}$ and $\vec{r}_{q_i} = \vec{R}_{s_i}$, to obtain:

$$P_{\text{bound}}(\text{orn}, n) \simeq \frac{1}{Z}\frac{V_1^{n+1}}{V_c}\prod_{i=1}^{n} f(|s_i-s_{i-1}|\,d, \sqrt{q_i-q_{i-1}}\,l_0)e^{E_{\text{DBD}}+nE_B} = \frac{1}{Z}r_{\text{DBD}}\prod_{i=1}^{n}\mathcal{P}(E_B, r_{\text{orn},i}, l_{\text{orn},i}), \tag{S7}$$

Where $r_{\text{DBD}} \equiv \frac{V_1}{V_c}e^{E_{\text{DBD}}}$, $r_{\text{orn},i} = |s_i - s_{i-1}|\,d$, $l_{\text{orn},i} = \sqrt{q_i-q_{i-1}}\,l_0$, and $\mathcal{P}(E_B, r_{\text{orn},i}, l_{\text{orn},i}) = e^{E_B}V_1 f(r_{\text{orn},i}, l_{\text{orn},i})$.

To give a specific example, consider the binding topology corresponding to **Fig. S2**. In this case, $n = 3$, $l_{\text{orn},1} = l_0$ (since there is one spring between the DBD and the first binding site), $l_{\text{orn},2} = \sqrt{3}l_0$ (since there are 3 springs between q_1 and q_2) and similarly $l_{\text{orn},3} = l_0$. **Eq. [S6]** implies that the contribution of this diagram to the binding probability is: $\frac{1}{Z}\frac{V_1^4}{V_c}\frac{27}{\left(2\pi l_0^2\right)^{9/2}}e^{3E_B+E_{\mathrm{DBD}}-5d^2/l_0^2}$. Note that while in this example, the order of the target sites is monotonic (i.e., as we increase the index of the binding sites, we progressively move further away from the DBD), in principle, we should sum over *all* diagrams, including those where the order is non-monotonic. At $n = 0$, P_{bound} (orn, 0) = $\frac{1}{Z}r_{\mathrm{DBD}}$.

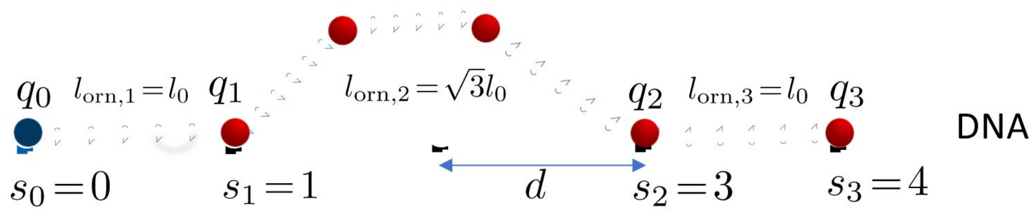

Figure S2

A diagram of the bound configurations at $n = 3$.

Similarly, for the state where the DBD is unbound and $n \geq 2$ IDR sites are bound, the binding probability reads:

$$P_{\mathrm{unbound}}(\mathrm{orn}, n) = \frac{1}{Z}\int_{\vec{r}_0\notin V_1} d^3\vec{r}_0 \prod_{\ell=1}^{n}\int_{\vec{r}_{q_\ell}\in V_1} d^3\vec{r}_{q\ell}\prod_{\ell=1}^{n_b-n}\int_{\vec{r}_{\bar{q}_\ell}\notin V_1} d^3\vec{r}_{\bar{q}\ell}\,\rho(\vec{r}_0,\ldots,\vec{r}_{n_b})e^{nE_B} \tag{S8}$$

$$\simeq \frac{1}{Z}r_{\mathrm{IDR}}\prod_{i=2}^{n}\mathcal{P}(E_B, r_{\mathrm{orn},i}, l_{\mathrm{orn},i}), \tag{S9}$$

where $r_{\mathrm{IDR}} \equiv \frac{V_1}{V_c}e^{E_B}$. For $n = 1$, Punbound (orn, 1) = $\frac{n_t n_b}{Z}r_{\mathrm{IDR}}$. Here $n_t n_b$ represents the number of possibilities for one of the n_b IDR sites bound to one of the n_t targets. For $n = 0$, P_{unbound} (orn, 0) = $\frac{1}{Z}\prod_{i=0}^{n_b}\int_{\vec{r}_i\notin V_1} d^3\vec{r}_i\,\rho(\vec{r}_0,\ldots,\vec{r}_{n_b}) \simeq \frac{1}{Z}$.

Since the sum of all the possibilities is equal to 1, as given by: $\sum_{n=0}^{n_{\min}}\sum_{\mathrm{orn}}\left(P_{\mathrm{bound}}+P_{\mathrm{unbound}}\right)=1$, where $n_{\min} = \min\{n_b, n_t\}$, we can determine the normalization factor. Therefore, the binding probability P_{TF} for the DBD bound in its target can be expressed as:

$$P_{\mathrm{TF}} = \frac{r_{\mathrm{DBD}}\left(1+\sum_{n=1}^{n_{\min}}\sum_{\mathrm{orn}}\prod_{i=1}^{n}\mathcal{P}(E_B, r_{\mathrm{orn},i}, l_{\mathrm{orn},i})\right)}{r_{\mathrm{DBD}}\left(1+\sum_{n=1}^{n_{\min}}\sum_{\mathrm{orn}}\prod_{i=1}^{n}\mathcal{P}(E_B, r_{\mathrm{orn},i}, l_{\mathrm{orn},i})\right)+1+r_{\mathrm{IDR}}\left(n_t n_b+\sum_{n=2}^{n_{\min}}\sum_{\mathrm{orn}}\prod_{i=2}^{n}\mathcal{P}(E_B, r_{\mathrm{orn},i}, l_{\mathrm{orn},i})\right)}. \tag{S10}$$

When $P_{TF} \ll 1$, we can approximate P_{TF} as

$$P_{\mathrm{TF}} \approx \left(1+\sum_{n=1}^{n_{\min}}\sum_{\mathrm{orn}}\prod_{i=1}^{n}\mathcal{P}(E_B, r_{\mathrm{orn},i}, l_{\mathrm{orn},i})\right)P_{\mathrm{simple}}, \tag{S11}$$

where we have considered the facts that $r_{\mathrm{IDR}} < r_{\mathrm{DBD}}$ because $E_B < E_{\mathrm{DBD}}$, and $P_{\mathrm{simple}} = r_{\mathrm{DBD}}/(1 + r_{\mathrm{DBD}}) \approx r_{\mathrm{DBD}}$ due to $r_{\mathrm{DBD}} \approx 1$. For instance, taking $E_{\mathrm{DBD}} = 15$, $V_1 = 1\mathrm{bp}^3 \approx (0.34\mathrm{nm})^3$, and $V_c = 1\mu\mathrm{m}^3$, we find that $r_{\mathrm{DBD}} \sim 10^{-4}$.

At $n_b = n_t = 1$, the generalized **Eq. [S10]** simplifies to the following expression:

$$P_{\mathrm{TF}} = \frac{r_{\mathrm{DBD}}\left[1 + e^{E_B} V_1 f(d, l_0)\right]}{r_{\mathrm{DBD}}\left[1 + e^{E_B} V_1 f(d, l_0)\right] + r_{\mathrm{IDR}} + 1}. \quad \text{(S12)}$$

Therefore, P_{TF} can be approximated as $(1+\mathscr{P})P_{\mathrm{simple}}$, as illustrated in **Eq. [6]** of the main text. This approximation is also derivable from **Eq. [S11]**.

S3. Q varying with l_0/d for all combinations of n_b and n_t

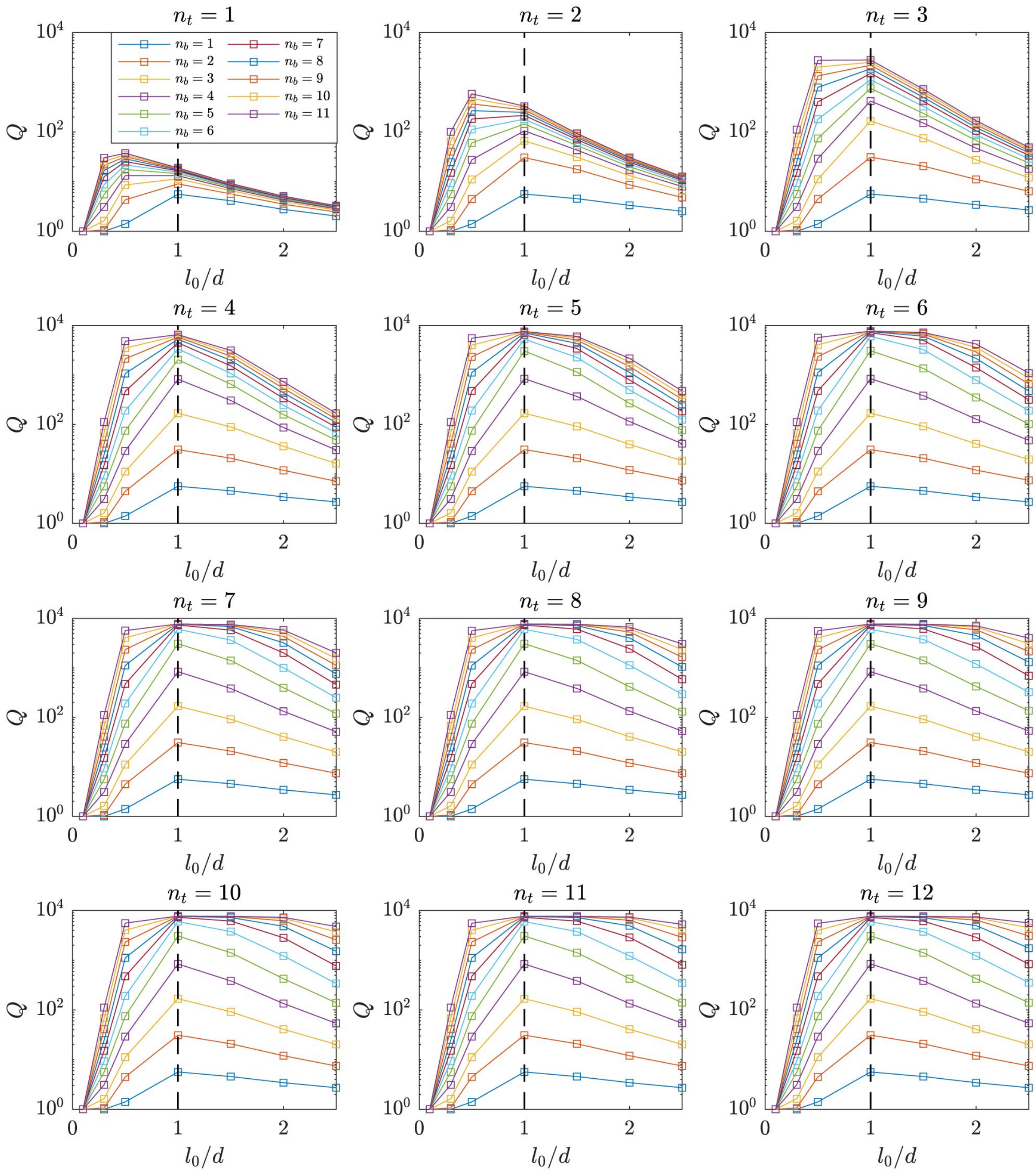

Figure S3

Maximal Q is observed at $l_0 \sim d$ for all combinations of n_b and n_t.

S4. P_{TF} varying with n_b and n_t

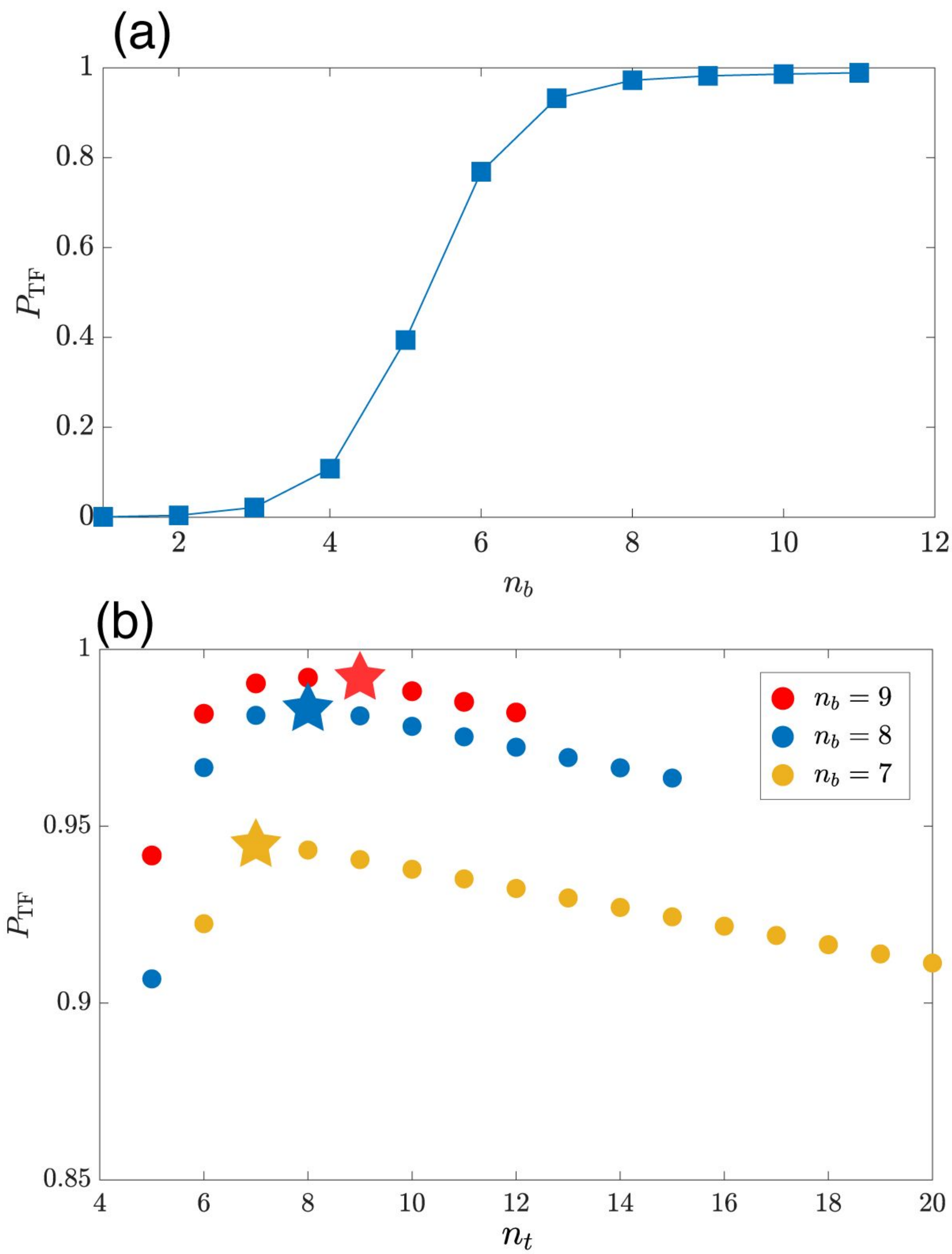

Figure S4

(a) P_{TF} varying with n_b at n_t = 12. P_{TF} initially increases rapidly, until eventually saturating. (b) P_{TF} varies non-monotonically with n_t. P_{TF} slightly decreases with increasing n_t at $n_t > n_b$. Three examples at n_b = 7, 8, 9 are displayed. P_{TF} is denoted by pentagrams at $n_t = n_b$.

S5. Robustness of the relationships: $n_b^* \sim n_t^*$ and n_b^* decreases with increasing E_B

Fig. S5 shows the contour lines at P_{TF} =0.9 and P_{TF} = 0.5 for various E_B. All take an approximate symmetrical L-shape, indicating that for a given E_B, the optimal values n_b^* and n_t^* are comparable, as guided by the black dashed line $y = x$. As E_B increases, the contour lines shift towards lower values of n_b and n_t, indicating a decrease in n_b^* as E_B increases.

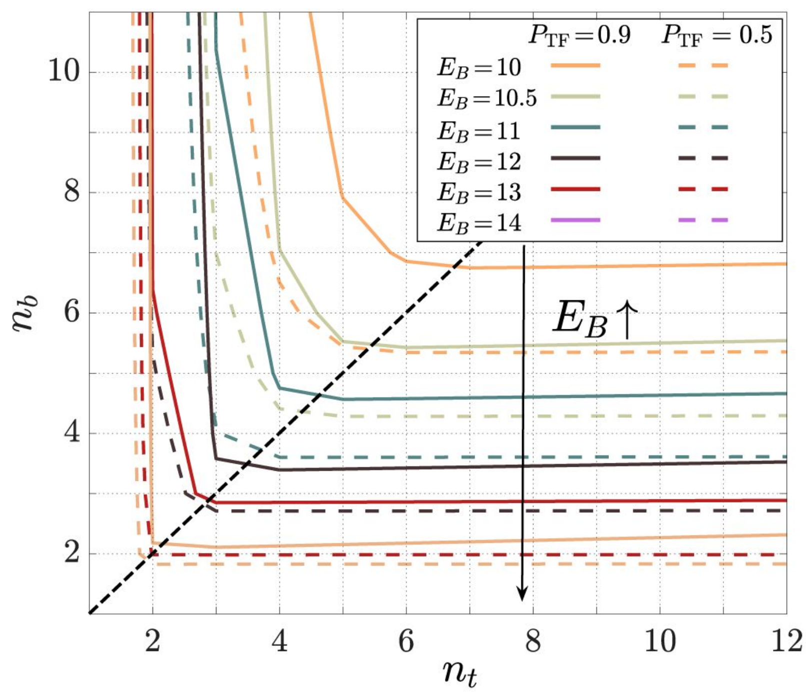

Figure S5

Contour lines at P_{TF} =0.9 and P_{TF} =0.5 on the plane defined by the number of binding sites, n_b, and the number of binding targets, n_t.

The black dashed line is added to guide the relationship $n_b = n_t$.

S6. Robustness of the binding design principles

In the main text, we employed uniform binding site distances in TFs, represented as l_0, and uniform binding target distances on the DNA, denoted as d, to obtain the design principles. Here, we also explore the scenario where both IDR binding sites and their targets are distributed as a Poisson process (i.e., the distance between neighboring sites is exponentially distributed), with mean values of l_0 and d, respectively. **Figure S6** is similar to **Figure 2** in the main text. We observe that the design principles identified in the main text are applicable under these conditions as well.

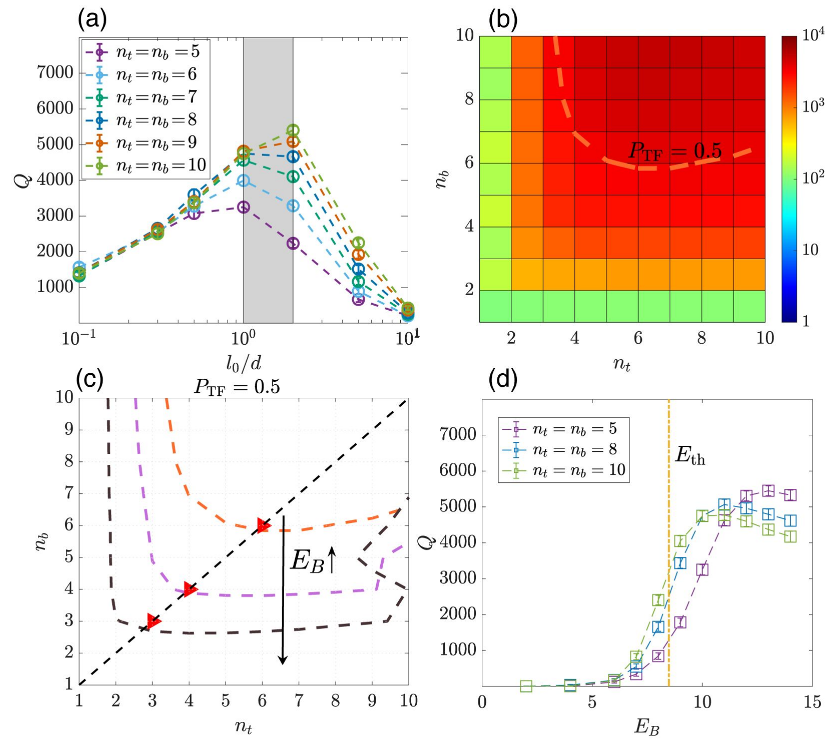

Figure S6

Plots of the affinity enhancement, quantified by the ratio $Q = P_{\mathrm{TF}}/P_{\mathrm{simple}}$, for the case where IDR sites as well as their targets are randomly positioned on the IDR and DNA, respectively.

See the analogous **Figure 2** in the main text for the case of homogeneously distributed IDR sites and targets. (a) Q is maximal when $l_0 \sim d$ (shaded region) for different values of $n_t = n_b$. The average distance between IDR sites is l_0, and the average distance between IDR targets is d. (b) A heatmap of Q at $E_B = 10$ and l_0 = d. (c) Contour lines at different E_B at $P_{\mathrm{TF}} = 0.5$ exhibit an approximate symmetrical L-shape, which indicates $n_b^* = n_t^*$, as shown by the red triangles. n_b^* decreases with increasing E_B. (d) Q varies with E_B for several different constants of $n_t = n_b$. The characteristic energy E_{th}, as predicted by **Eq. [5]** in the main text, corresponds to the region where a notable increase is observed.

S7. TF walking along the antenna once it attaches, and t_{total} vs. E_B and vs. $\tilde{n}$

We show a typical example of the trajectory of a TF (red) in the cell nucleus (light blue) in **Fig. S7(a)** where the TF walks along the antenna to the DBD target once it attaches the IDR targets at $\tilde{n}^*$ and E_B^*. The DBD target is at the origin. In **Fig. S7(b)**, we find that the average number of rounds the TF hits the antenna, n_{hits}, is close to 1, so it is fair to describe the overall search process as two distinct steps with the first step being purely a 3D search and the second a purely 1D search. This is expressed by **Eq. [5]** in the main text.

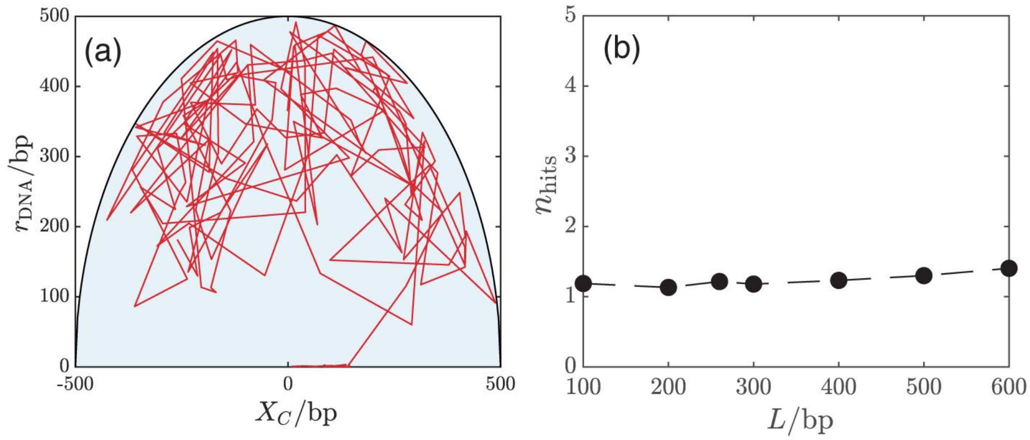

Figure S7

(a) A trajectory (red) of TF in the cell nucleus (light blue). r_{DNA} is the average distance of the TF to the antenna, and X_c is the position of the center of mass along the antenna direction. The trajectory is displayed with a time interval of $10^6 t_0$. (b) The average number of rounds the TF hits the antenna, n_{hits}, before reaching the DBD target. The optimal mean total search time is obtained at $L^* = 300$bp.

The minimal search time, t_{min}, is obtained at L^*, E_B^*, and $\tilde{n}^*$. **Fig. S8(a)** shows t_{total} vs. E_B at fixed L^* and $\tilde{n}^*$. When $E_B > E_B^*$, t_{3D} does not depend on E_B as we predict (red solid line) and t_{1D} exponentially increases with increasing E_B. The solid blue line shows the exponential dependence of t_{1D} on E_B, also shown in the inset. When $E_B < E_B^*$, the number of rounds of hits on the antenna, n_{hits}, is larger than 1, as displayed in the inset. In this scenario, **equation (5)** in the main text becomes ineffective, resulting in a significantly longer search time than t_{min}.

Fig. S8(b) shows t_{total} vs. $\tilde{n}$ at fixed L^* and E_B^*. When $\tilde{n} > \tilde{n}^*$, t_{3D} and t_{1D} agree with the prediction of **Eq. [5]** (red curve and blue curve). Interestingly, we find that t_{1D} only weakly depends on $\tilde{n}$. This arises due to the compensating effects of attaching and detaching: As $\tilde{n}$ increases, the TF attaches to new targets more easily, speeding up 1D motion. However, it also makes detaching from the current site more difficult, thereby slowing down 1D motion. When $\tilde{n} < \tilde{n}^*$, **Eq. [5]** breaks down because of multiple rounds of hits, which is verified in the inset. As a result, both t_{total} and t_{3D} increases with decreasing $\tilde{n}$, leading them to become significantly longer than t_{min}. If each round of 3D search is independent, the 3D search time should be n_{hits} times the one round of 3D search time, displayed in the black curve.

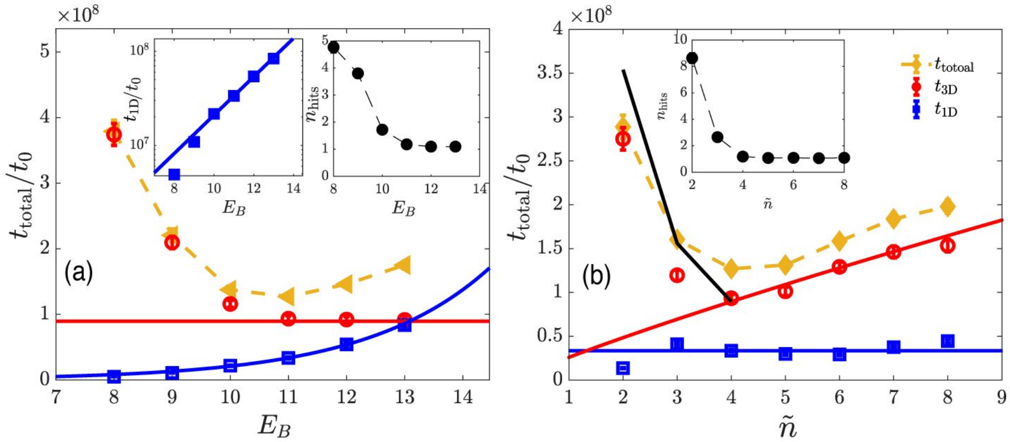

Figure S8

(a) t_{total}, t_{3D}, and t_{1D} vs. E_B at fixed L^* and $\tilde{n}^*$. The red solid line is the prediction of t_{3D} from **Eq. [5]**, and the blue line (also in the inset) is the exponential fit for t_{1D}. The black curve in the inset is the number of rounds the TF hits the antenna, n_{hits}, vs. E_B. (b) t_{total}, t_{3D}, and t_{1D} vs. $\tilde{n}$ at fixed L^* and E_B^*. The red and blue solid lines represent the predictions of t_{3D} and t_{1D} from **Eq. [5]**, respectively. n_{hits} varying with $\tilde{n}$ is shown in the inset.

S8. A single round of 3D-1D search is typically the case in the optimal search process

In this section, we will show that the optimal search process (i.e., minimizing the mean first passage time) generically takes place via a single round of 3D-1D search. To do so, we first need to derive the expression for the mean total search time, t_{total}, and its dependence on the relevant parameters, including, importantly, E_B and L. According to our model presented in the main text: A TF is initially in the 3D nucleus space; it can attach to the antenna and perform 1D diffusion with the coefficient denoted by D_1. We assume the DBD target is placed in the middle of the antenna, as shown in **Fig. 1** of the main text. The mean search time taken for a TF to move from 3D space and attach to the antenna is denoted by t_{3D}. The TF can also detach from the antenna at a rate we denote by Γ, perform a 3D random walk, until ultimately attaching again to the antenna. In the following, we will make the assumption (often made in the context of FD model [23–29]) that the binding position after such a 3D excursion is uniformly distributed on the antenna. Both D_1 and Γ are functions of E_B. When the TF is on the antenna, its mean search time to reach the DBD target, located a distance x away on the antenna, is represented by $\tau(x)$. Naturally, $\tau(x)$ is zero when $x = 0$, the location of the DBD target. The mean total search time t_{total} is thus given by

$$t_{\mathrm{total}} = \frac{1}{L}\int_{-L/2}^{L/2} \mathcal{T}(x)dx + t_{\mathrm{3D}}. \tag{S13}$$

Consider a time interval Δt. With probability $\Gamma\Delta t$ the TF falls off and the mean first passage time to reach the target is $t_{\text{total}} + \Delta t$. Otherwise, with probability $1 - \Gamma\Delta t$, the mean first passage time would be the average of the two possible outcomes, namely, $(\tau(x + \Delta t) + \tau(x + \Delta t)/2 + \Delta t$. This recursive reasoning leads us to the following equation:

$$\mathcal{T}(x) = \Delta t + \frac{\mathcal{T}(x - \Delta x) + \mathcal{T}(x + \Delta x)}{2}(1 - \Gamma\Delta t) + t_{\text{total}}\Gamma\Delta t. \tag{S14}$$

In the continuous limit, we have

$$D_1\frac{\partial^2\mathcal{T}(x)}{\partial x^2} - \Gamma\mathcal{T}(x) + \Gamma t_{\text{total}} + 1 = 0, \tag{S15}$$

where $D_1 = (\Delta x)^2/(2\Delta t)$. By applying the boundary conditions, specifically the absorbing boundary condition at $x = 0$ and the reflecting boundary conditions at $x = \pm L/2$, $\tau(x)$ is determined as

$$\mathcal{T}(x) = \left(t_{\text{total}} + \Gamma^{-1}\right)\left[1 - \frac{\cosh(\frac{L-2|x|}{L_s})}{\cosh(\frac{L}{L_s})}\right], \tag{S16}$$

where L_s, representing the 1D diffusion distance, is defined by $L_s \equiv 2\sqrt{D_1\Gamma^{-1}}$ [59]. Therefore, t_{total} can be self-consistently solved by substituting $\tau(x)$ into Eq. [S13], presented as follows:

$$t_{\text{total}} = \frac{L}{L_s}\coth\left(\frac{L}{L_s}\right)t_{3\text{D}} + \left[\frac{L}{L_s}\coth\left(\frac{L}{L_s}\right) - 1\right]\Gamma^{-1}. \tag{S17}$$

We can obtain t_{total} in two limiting cases:

- When $L \gg L_s$, where multiple rounds of 3D-1D excursions are needed to find the target, we have $t_{\text{total}} = \frac{L}{L_s}\left(t_{3\text{D}} + \Gamma^{-1}\right)$. This is the same as the results of the facilitated diffusion model [28, 29, 59, 60].
- When $L \ll L_s$, where a single round of 3D-1D search can find the target, we get $t_{\text{total}} = t_{3\text{D}} + \frac{L^2}{12D_1}$. This matches Eq. [5] in the main text when $L < 2R$.

Note that the calculations above did not assume an optimal or efficient search process, but simply characterized the dependence of the mean first passage time on the model parameters. Next, based on the expression for t_{total} in Eq. [S17], we will prove that the search process involving multiple rounds of 3D-1D search is *not* optimal. This conclusion is reached by showing that the minimum t_{total} always occurs in the regime where $L < 2L_s$ (since L_s is the typical distance covered by 1D diffusion on the antenna before falling off, and the maximal distance needed to be traversed on the antenna is $L/2$). We emphasize an important distinction between our model and the well-studied facilitated diffusion model (where it is established that *multiple* rounds of search are needed for efficient search): in our case the antenna length L is a *variable* that can be optimized over (since we assume evolution has optimized over the size of the region flanking the DBD in which binding sites for the IDR are located). In the facilitated diffusion model, L is assumed to be the length of the genome, since the interactions of the TF and the DNA are assumed non-specific.

To proceed, let us first note that the factor coth $\left(\frac{L}{L_s}\right)$ in Eq. [S17] quickly converges to 1 as the ratio of L/L_s increases. For instance, when the ratio is 2, we have coth(2) ≈ 1.04. Thus, for $L > 2L_s$, we could approximate t_{total} as

$$t_{\text{total}} \approx \frac{L}{L_s} t_{\text{3D}} + \left(\frac{L}{L_s} - 1\right) \Gamma^{-1}. \tag{S18}$$

Then, we consider how t_{3D} varies with L. If the rate to hit the antenna after 3D diffusion was the sum of the rates to hit every infinitesimal segment of it, we would expect the rate to hit the antenna to be *linear* in L, and the mean first passage time to scale as $1/L$. However, these processes are not uncorrelated. Interestingly, when $L < 2R$, and assuming a straight antenna, t_{3D} is proportional to $\ln(L)/L$ (see Eq. [5] in the main text), implying that for a straight antenna the effect of these correlations is mild. However, when $L > 2R$, the antenna is curved rather than straight (due to the limitations imposed on the DNA conformation by the dimensions of the nucleus), and the dependence of t_{3D} on L is weaker [61]. Thus, for any given L greater than $2L_s$, t_{3D} varies more weakly than $1/L$, making the first term of t_{total} in Eq. [S18] an increasing function of L. Clearly, the second term increases as well. Therefore, when $L > 2L_s$, regardless of the functional forms of D_1 and Γ^{-1} on E_B, t_{total} decreases with decreasing L. Thus, the minimum ttotai occurs in the regime where $L < 2L_s$, and the optimal search typically consists of a single search round.

S9. Robustness of mean total search time with changing the antenna geometry

We checked that the mean total search time (Mean First Passage Time) for IDR targets, generated via the worm-like chain model - a random walk with a persistence length [62], does not significantly differ from the scenario where targets are arranged linearly, as presented in the main text. See **Fig. S9** for details.

S10. MSD in a complex DNA configuration

In **Fig. S10**, we run simulations to examine how the Mean Square Displacement (MSD) of a point searcher is influenced by the binding energy (E_{nonsp}) between the TF and the entire DNA in a complex DNA configuration. **Fig. S10(a)** shows the DNA configuration generated via the worm-like chain model with a persistence length of 150 bp [62], where the volume fraction $\nu = 0.008$ is comparable to the experimental yeast DNA volume fraction. **Fig. S10(b)** shows the MSD measured at different values of E_{nonsp} for $\nu = 0.008$. When $E_{\text{nonsp}} \sim 1k_BT$, which corresponds to the non-specific binding energy regime in experiments [63], the effective 3D diffusion coefficient is slightly modified, which does not affect our conclusion on the optimal search process. Additionally, we observed that the MSD curve bends when E_{nonsp} exceeds the typical range of non-specific binding energies.

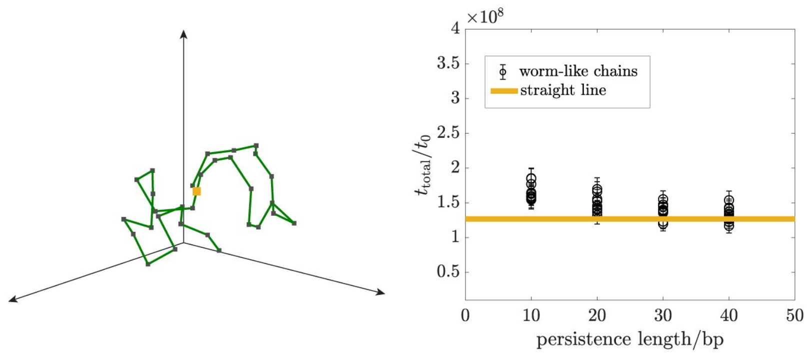

Figure S9

Left: an example of the target configuration (the antenna) generated by the worm-like chain model, where *L* = 300bp, *d* = 10bp, and the persistence length also equals 10bp.

Right: Mean total search times weakly vary with persistence length when targets are generated using the worm-like chain model. The values are represented by the black dots, with parameters set to $E_B = 11$, $L = 300$ bp, and $\tilde{n} = 4$. For each persistence length, we obtained ten mean total search times by considering ten different target configurations. The mean total search times do not exhibit significant differences compared to that for targets arranged linearly (indicated by the horizontal orange line).

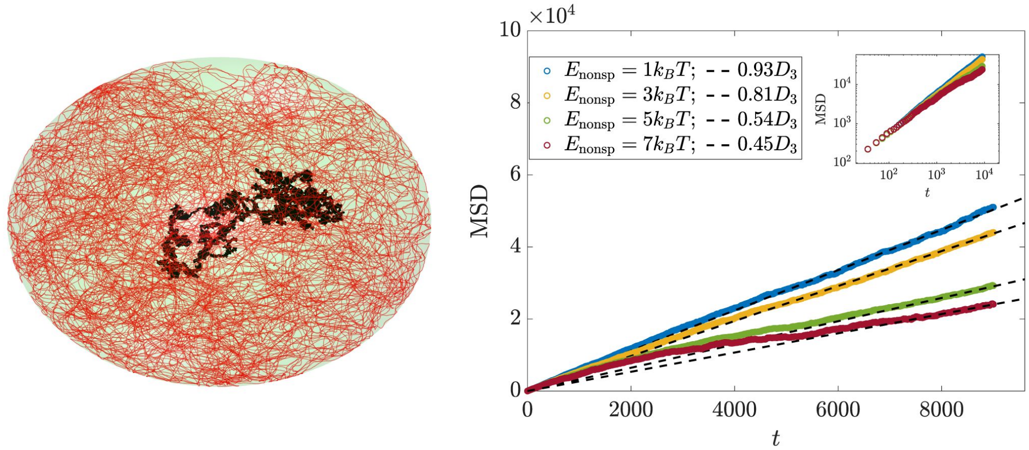

Figure S10

(a) Configuration of DNA (in red) at v = 0.008 with a complex structure and a trajectory of TF (in black). A TF acts as a point searcher that diffuses in 3D space and becomes trapped once it hits the DNA, which is 1nm wide. It detaches at a rate of $\frac{D_3}{1nm^2}\exp(-E_{\text{nonsp}}/k_BT)$. (b) MSD at different E_{nonsp}.

S11. Properties of the octopusing walk

Once an IDR site binds to the antenna, the TF performs an octopusing walk, whereby different IDR binding sites constantly bind and unbind from the DNA. As we shall see, this will lead to an effective 1D diffusion of the TF. This indicates that during the octopusing walk, the TF neither gets trapped on a single target nor fully detaches from the antenna, allowing for rapid dynamics (in fact, the 1D diffusion coefficient can be comparable to the free 3D diffusion coefficient). We shall also show that the number of bound sites on the DNA is described by a probability distribution that is in steady-state and obeys detailed balance.

We denote that time-dependent probability distribution of the number of bound sites by $P(n, t)$ (with n ranging from 0 to $\tilde{n}$). The dynamics of this probability distribution is given by:

$$\frac{dP(n,t)}{dt} = P(n-1,t)W_{n-1,n} + P(n+1,t)W_{n+1,n} + P(n,t)W_{n,n}, \tag{S19}$$

where W is called *transition rate matrix*, and $W_{n,n} = -W_{n,n-1} - W_{n,n+1}$. Intuitively, we anticipate that $W_{n,n-1}$, corresponding to the process of *detachment* of one of the sites, is *linear* in n, since we expect the different sites to be approximately independent of each other. Similarly, if we consider the different IDR sites as independent “searchers”, the on-rate (corresponding to $W_{n,n+1}$) to be linear in ($\tilde{n}$ — n). Indeed, **Fig. S11(a)** corroborates this intuition to an excellent approximation. As the dynamics proceeds, **Eq. [S19]** would reach a steady-state distribution governed by the matrix W. **Fig. S11(b)** shows the excellent agreement between the steady-state associated with the matrix W and that found directly from the MD simulation. This confirms that TF movement along the antenna is in a dynamic steady state. We also note that since our transition matrix is tri-diagonal, it satisfies the detailed balance condition, i.e., in the steady-state reached, we have: $P(n)W_{n,n-1} = P(n-1)W_{n-1,n}$, where $P(n)$ is the steady-state distribution.

A *priori*, one may think that the TF is facing two contradictory demands: on the one hand, it has to perform rapid diffusion along the DNA. On the other hand, it needs to bind sufficiently strongly so that it does not fall off the DNA too soon. Indeed, this is known as the speed-stability paradox within the context of TF search in bacteria. Importantly, the simple picture we described above explains an elegant mechanism to bypass this paradox, achieving a high diffusion coefficient with a low rate of detaching from the DNA. This comes about, due to the compensatory nature of the dynamics outlined above: there is, in effect, a feedback mechanism whereby if only few sites are attached to the DNA, the on-rate increases, and, similarly, if too many are attached (thus prohibiting the 1D diffusion) the off-rate increases. This is manifested by the peaked steady-state distribution shown in **Fig. S11(b)**. In fact, this mechanism is remarkably effective even when the typical (i.e., most probable) number of binding sites is as small as 2! (note that in this case, as shown in the inset of **Fig. S11(a)**, the TF may cover a distance of more than 100bp before falling off).

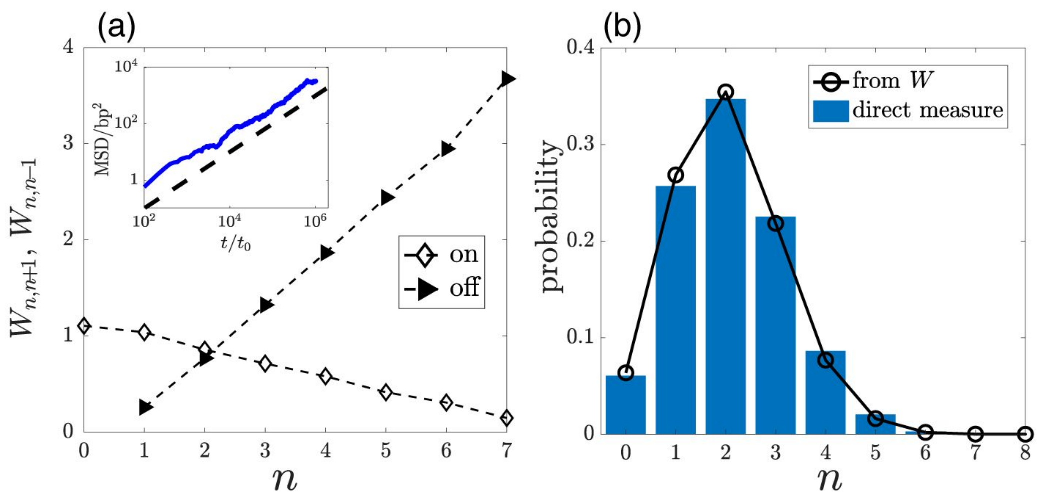

Figure S11

(a) $W_{n,n+1}$ (or $W_{n,n-1}$) denoted by “on” (or “off”) varies with n for E_B = 6 and ö = 12. Inset: Mean square displacement (MSD) suggests diffusive motion, as indicated by the dashed line following MSD ∞ t. (b) Steady probability distribution of bound sites.

References

[1] Latchman David S. (1997) **Transcription factors: An overview. The International Journal of Biochemistry** *Cell Biology* **29**:1305–1312 https://doi.org/10.1016/s1357-2725(97)00085-x | Google Scholar

[2] Mitchell Pamela J., Tjian Robert (1989) **Transcriptional regulation in mammalian cells by sequence-specific dna binding proteins** *Science* **245**:371–378 https://doi.org/10.1126/science.2667136 | Google Scholar

[3] Ptashne Mark, Gann Alexander (1997) **Transcriptional activation by recruitment** *Nature* **386**:569–577 https://doi.org/10.1038/386569a0 | Google Scholar

[4] Lambert Samuel A., Jolma Arttu, Campitelli Laura F., Das Pratyush K., Yin Yimeng, Albu Mihai, Chen Xiaoting, Taipale Jussi, Hughes Timothy R., Weirauch Matthew T. (2018) **The human transcription factors** *Cell* **172**:650–665 https://doi.org/10.1016/j.cell.2018.01.029 | Google Scholar

[5] Stadhouders Ralph, Filion Guillaume J., Graf Thomas (2019) **Transcription factors and 3d genome conformation in cell-fate decisions** *Nature* **569**:345–354 https://doi.org/10.1038/s41586-019-1182-7 | Google Scholar

[6] Jacob François, Monod Jacques (1961) **Genetic regulatory mechanisms in the synthesis of proteins** *Journal of Molecular Biology* **3**:318–356 https://doi.org/10.1016/s0022-2836(61)80072-7 | Google Scholar

[7] Ferrie John J, Karr Jonathan P, Tjian Robert, Darzacq Xavier (2022) **"Structure"-function relationships in eukaryotic transcription factors: The role of intrinsically disordered regions in gene regulation** *Molecular Cell* https://doi.org/10.1016/j.molcel.2022.09.021 | Google Scholar

[8] Jonas Felix, Navon Yoav, Barkai Naama (2025) **Intrinsically disordered regions as facilitators of the transcription factor target search** *Nature Reviews Genetics* **26**:424–435 https://doi.org/10.1038/s41576-025-00816-3 | Google Scholar

[9] Liu Jiangang, Perumal Narayanan B., Oldfield Christopher J., Su Eric W., Uversky Vladimir N., Dunker A. Keith (2006) **Intrinsic disorder in transcription factors** *Biochemistry* **45**:6873–6888 https://doi.org/10.1021/bi0602718 | Google Scholar

[10] Fuxreiter Monika, Simon Istvan, Bondos Sarah (2011) **Dynamic protein-DNA recognition: beyond what can be seen** *Trends in Biochemical Sciences* **36**:415–423 https://doi.org/10.1016/j.tibs.2011.04.006 | Google Scholar

[11] Guo Xin, Bulyk Martha L, Hartemink Alexander J (2012) **Intrinsic disorder within and flanking the dna-binding domains of human transcription factors** *Biocomputing* World Scientific :104–115 https://doi.org/10.1142/9789814366496_0011 | Google Scholar

[12] Wright Peter E., Dyson H. Jane (2014) **Intrinsically disordered proteins in cellular signalling and regulation** *Nature Reviews Molecular Cell Biology* **16**:18–29 https://doi.org/10.1038/nrm3920 | Google Scholar

[13] Garcia Daniel A., Chen Wei-Ting, Ren Shuying, Chen Long-Chi, Chen Xiaonan, Yu Shiqi, Han Zhijie, Kalugin Sergei, Kono Ken, Noberini Roberta, et al. (2021) **An intrinsically disordered region-mediated confinement state contributes to the dynamics and function of transcription factors** *Molecular Cell* **81**:1484–1498 https://doi.org/10.1016/j.molcel.2021.01.013 | Google Scholar

[14] Brown Celeste J., Johnson Audra K., Daughdrill Gary W. (2010) **Comparing models of evolution for ordered and disordered proteins** *Molecular Biology and Evolution* **27**:609–621 https://doi.org/10.1093/molbev/msp277 | Google Scholar

[15] Zarin Taraneh, Strome Bob, Ba Alex N Nguyen, Alberti Simon, Forman-Kay Julie D, Moses Alan M (2019) **Proteome-wide signatures of function in highly diverged intrinsically disordered regions** *eLife* **8**:e46883 https://doi.org/10.7554/eLife.46883 | Google Scholar

[16] Vuzman Dana, Levy Yaakov (2010) **Dna search efficiency is modulated by charge composition and distribution in the intrinsically disordered tail** *Proceedings of the National Academy of Sciences* **107**:21004–21009 https://doi.org/10.1073/pnas.1011775107 | Google Scholar

[17] Vuzman Dana, Levy Yaakov (2012) **Intrinsically disordered regions as affinity tuners in protein-dna interactions** *Molecular Biosystems* **8**:47–57 https://doi.org/10.1039/c1mb05273j | Google Scholar

[18] Brodsky Sagie, Jana Tamar, Mittelman Karin, Chapal Michal, Kumar Divya Krishna, Carmi Miri, Barkai Naama (2020) **Intrinsically disordered regions direct transcription factor in vivo binding specificity** *Molecular Cell* **79**:459–471.e4 https://doi.org/10.1016/j.molcel.2020.05.032 | Google Scholar

[19] Kumar Divya Krishna, Jonas Felix, Jana Tamar, Brodsky Sagie, Carmi Miri, Barkai Naama (2023) **Complementary strategies for directing in vivo transcription factor binding through DNA binding domains and intrinsically disordered regions** *Molecular Cell* **83**:1462–1473.e5 https://doi.org/10.1016/j.molcel.2023.04.002 | Google Scholar

[20] Iwahara Junji, Zweckstetter Markus, Clore G. Marius (2006) **Nmr structural and kinetic characterization of a homeodomain diffusing and hopping on nonspecific dna** *Proceedings of the National Academy of Sciences* **103**:15062–15067 https://doi.org/10.1073/pnas.0605868103 | Google Scholar

[21] Boija A., Klein I. A., Sabari B. R., Dall'Agnese A., Coffey E. L., Zamudio A. V., Li C. H., Shrinivas K., Manteiga J. C., Hannett N. M., et al. (2018) **Transcription factors activate genes through the phase-separation capacity of their activation domains** *Cell* **175**:1842–1855 https://doi.org/10.1016/j.cell.2018.10.042 | Google Scholar

[22] Chappleboim Michal, Naveh-Tassa Segev, Carmi Miri, Levy Yaakov, Barkai Naama (2024) **Ordered and disordered regions of the origin recognition complex direct differential in vivo binding at distinct motif sequences** *Nucleic Acids Research* **52**:5720–5731 https://doi.org/10.1093/nar/gkae249 | Google Scholar

[23] Berg Otto G, Winter Robert B, Hippel Peter H Von (1981) **Diffusion-driven mechanisms of protein translocation on nucleic acids. 1. models and theory** *Biochemistry* **20**:6929–6948 https://doi.org/10.1021/bi00527a028 | Google Scholar

[24] Berg Otto G., Hippel Peter H. von (1987) **Selection of DNA binding sites by regulatory proteins** *Journal of Molecular Biology* **193**:723–743 https://doi.org/10.1016/0022-2836(87)90354

-8 | Google Scholar

[25] Slutsky Michael, Mirny Leonid A. (2004) **Kinetics of protein-DNA interaction: Facilitated target location in sequence-dependent potential** *Biophysical Journal* **87**:4021–4035 https://doi.org/10.1529/biophysj.104.050765 | Google Scholar

[26] Mirny Leonid, Slutsky Michael, Wunderlich Zeba, Tafvizi Anahita, Leith Jason, Kosmrlj Andrej (2009) **How a protein searches for its site on dna: the mechanism of facilitated diffusion** *Journal of Physics A: Mathematical and Theoretical* **42**:434013 https://doi.org/10.1088/1751-8113/42/43/434013 | Google Scholar

[27] Li Gene-Wei, Berg Otto G., Elf Johan (2009) **Effects of macromolecular crowding and DNA looping on gene regulation kinetics** *Nature Physics* **5**:294–297 https://doi.org/10.1038/nphys1222 | Google Scholar

[28] Benichou Olivier, Loverdo Claude, Moreau Michel, Voituriez Raphael (2011) **Intermittent search strategies** *Reviews of Modern Physics* **83**:81 https://doi.org/10.1103/RevModPhys.83.81 | Google Scholar

[29] Sheinman M, Benichou O, Kafri Y, Voituriez R (2012) **Classes of fast and specific search mechanisms for proteins on dna** *Reports on Progress in Physics* **75**:026601 https://doi.org/10.1088/0034-4885/75/2/026601 | Google Scholar

[30] Hammar Petter, Leroy Prune, Mahmutovic Anel, Marklund Erik G, Berg Otto G, Elf Johan (2012) **The lac repressor displays facilitated diffusion in living cells** *Science* **336**:1595–1598 https://doi.org/10.1126/science.1221648 | Google Scholar

[31] Dror Iris, Rohs Remo, Mandel-Gutfreund Yael (2016) **How motif environment influences transcription factor search dynamics: Finding a needle in a haystack** *BioEssays* **38**:605–612 https://doi.org/10.1002/bies.201600005 | Google Scholar

[32] Jana Tamar, Brodsky Sagie, Barkai Naama (2021) **Speed-specificity trade-offs in the transcription factors search for their genomic binding sites** *Trends in Genetics* **37**:421–432 https://doi.org/10.1016/j.tig.2020.12.001 | Google Scholar

[33] Wagh Kaustubh, Stavreva Diana A., Upadhyaya Arpita, Hager Gordon L. (2023) **Transcription factor dynamics: One molecule at a time** *Annual Review of Cell and Developmental Biology* **39** https://doi.org/10.1146/annurev-cellbio-022823-013847 | Google Scholar

[34] Danilowicz C., Kafri Y., Conroy R. S., Coljee V. W., Weeks J., Prentiss M. (2004) **Measurement of the phase diagram of DNA unzipping in the temperature-force plane** *Physical Review Letters* **93** https://doi.org/10.1103/physrevlett.93.078101 | Google Scholar

[35] Rouse Prince E (1953) **A theory of the linear viscoelastic properties of dilute solutions of coiling polymers** *The Journal of Chemical Physics* **21**:1272–1280 https://doi.org/10.1063/1.1699180 | Google Scholar

[36] Doi Masao, Edwards Samuel Frederick (1988) **The Theory of Polymer Dynamics** Oxford University Press Google Scholar

[37] Shimamoto Nobuo (1999) **One-dimensional diffusion of proteins along dna: its biological and chemical significance revealed by single-molecule measurements** *Journal of Biological Chemistry* **274**:15293–15296 https://doi.org/10.1074/jbc.274.22.15293 | Google Scholar

[38] Hu Tao, Shklovskii B. I. (2007) **Kinetics of viral self-assembly: Role of the single-stranded RNA antenna** *Physical Review E* **75** https://doi.org/10.1103/physreve.75.051901 | Google Scholar

[39] Castellanos Milagros, Mothi Nivin, Muñoz Victor (2020) **Eukaryotic transcription factors can track and control their target genes using DNA antennas** *Nature Communications* **11** https://doi.org/10.1038/s41467-019-14217-8 | Google Scholar

[40] Jonas Felix, Carmi Miri, Krupkin Beniamin, Steinberger Joseph, Brodsky Sagie, Jana Tamar, Barkai Naama (2023) **The molecular grammar of protein disorder guiding genome-binding locations** *Nucleic Acids Research* **51**:4831–4844 https://doi.org/10.1093/nar/gkad184 | Google Scholar

[41] Mindel Vladimir, Brodsky Sagie, Cohen Aileen, Manadre Wajd, Jonas Felix, Carmi Miri, Barkai Naama (2023) **Intrinsically disordered regions of the msn2 transcription factor encode multiple functions using interwoven sequence grammars** *Nucleic Acids Research* **52**:2260–2272 https://doi.org/10.1093/nar/gkad1191 | Google Scholar

[42] Marcovitz Amir, Levy Yaakov (2013) **Weak frustration regulates sliding and binding kinetics on rugged proteindna landscapes** *The Journal of Physical Chemistry B* **117**:13005–13014 https://doi.org/10.1021/jp402296d | Google Scholar

[43] Gennes PG De, Leger L (1982) **Dynamics of entangled polymer chains** *Annual Review of Physical Chemistry* **33**:49–61 https://doi.org/10.1146/annurev.pc.33.100182.000405 | Google Scholar

[44] Berg Howard C, Purcell Edward M (1977) **Physics of chemoreception** *Biophysical journal* **20**:193–219 https://doi.org/10.1016/S0006-3495(77)85544-6 | Google Scholar

[45] Berg Howard C (2018) **Random walks in biology** *Random Walks in Biology* Princeton University Press Google Scholar

[46] Guilbaud Sébastien, Salomé Laurence, Destainville Nicolas, Manghi Manoel, Tardin Catherine (2019) **Dependence of DNA persistence length on ionic strength and ion type** *Physical Review Letters* **122** https://doi.org/10.1103/physrevlett.122.028102 | Google Scholar

[47] Larson Daniel R., Zenklusen Daniel, Wu Bin, Chao Jeffrey A., Singer Robert H. (2011) **Real-time observation of transcription initiation and elongation on an endogenous yeast gene** *Science* **332**:475–478 https://doi.org/10.1126/science.1202142 | Google Scholar

[48] Hopfield J. J. (1974) **Kinetic proofreading: A new mechanism for reducing errors in biosynthetic processes requiring high specificity** *Proceedings of the National Academy of Sciences* **71**:4135–4139 https://doi.org/10.1073/pnas.71.10.4135 | Google Scholar

[49] Marklund Emil, Mao Guanzhong, Yuan Jinwen, Zikrin Spartak, Abdurakhmanov Eldar, Deindl Sebastian, Elf Johan (2022) **Sequence specificity in dna binding is mainly governed by association** *Science* **375**:442–445 https://doi.org/10.1126/science.abg7427 | Google Scholar

[50] Zentner Gabriel E., Kasinathan Sivakanthan, Xin Beibei, Rohs Remo, Henikoff Steven (2015) **ChEC-seq kinetics discriminates transcription factor binding sites by DNA sequence and

shape in vivo *Nature Communications* **6** https://doi.org/10.1038/ncomms9733 | Google Scholar

[51] Tenenbaum Debora, Inlow Koe, Friedman Larry J., Cai Anthony, Gelles Jeff, Kondev Jane (2023) **Rna polymerase sliding on dna can couple the transcription of nearby bacterial operons** *Proceedings of the National Academy of Sciences* **120** https://doi.org/10.1073/pnas.2301402120 | Google Scholar

[52] Einav Tal, Yazdi Shahrzad, Coey Aaron, Bjorkman Pamela J., Phillips Rob (2019) **Harnessing avidity: Quantifying the entropic and energetic effects of linker length and rigidity for multivalent binding of antibodies to hiv-1** *Cell Systems* **9**:466–474.e7 https://doi.org/10.1016/j.cels.2019.09.007 | Google Scholar

[53] Morgunova E., Taipale J. (2017) **Structural perspective of cooperative transcription factor binding** *Current Opinion in Structural Biology* **47**:1–8 https://doi.org/10.1016/j.sbi.2017.03.006 | Google Scholar

[54] Sabari B. R., Dall'Agnese A., Boija A., Klein I. A., Coffey E. L., Shrinivas K., Abraham B. J., Hannett N. M., Zamudio A. V., Manteiga J. C., et al. (2018) **Coactivator condensation at super-enhancers links phase separation and gene control** *Science* **361**:eaar3958 https://doi.org/10.1126/science.aar3958 | Google Scholar

[55] Kent Samantha, Brown Kyle, Yang Chou-hsun, Alsaihati Njood, Tian Christina, Wang Haobin, Ren Xiaojun (2020) **Phase-separated transcriptional condensates accelerate target-search process revealed by live-cell singlemolecule imaging** *Cell Reports* **33**:108248 https://doi.org/10.1016/j.celrep.2020.108248 | Google Scholar

[56] Holehouse Alex S., Kragelund Birthe B. (2023) **The molecular basis for cellular function of intrinsically disordered protein regions** *Nature Reviews Molecular Cell Biology* **25**:187–211 https://doi.org/10.1038/s41580-023-00673-0 | Google Scholar

[57] Weeks John D., Chandler David, Andersen Hans C. (1971) **Role of repulsive forces in determining the equilibrium structure of simple liquids** *The Journal of Chemical Physics* **54**:5237–5247 https://doi.org/10.1063/1.1674820 | Google Scholar

[58] Berthier Ludovic, Biroli Giulio (2011) **Theoretical perspective on the glass transition and amorphous materials** *Reviews of Modern Physics* **83**:587–645 https://doi.org/10.1103/revmodphys.83.587 | Google Scholar

[59] Wunderlich Zeba, Mirny Leonid A. (2008) **Spatial effects on the speed and reliability of protein-DNA search** *Nucleic Acids Research* **36**:3570–3578 https://doi.org/10.1093/nar/gkn173 | Google Scholar

[60] Hachmo Ori, Amir Ariel (2023) **Conditional probability as found in nature: Facilitated diffusion** *American Journal of Physics* **91**:653–658 https://doi.org/10.1119/5.0123866 | Google Scholar

[61] Tao Hu A., Yu Grosberg, Shklovskii B.I. (2006) **How proteins search for their specific sites on DNA: The role of DNA conformation** *Biophysical Journal* **90**:2731–2744 https://doi.org/10.1529/biophysj.105.078162 | Google Scholar

[62] Milstein Joshua N., Meiners Jens-Christian (2013) **Worm-Like Chain (WLC) Model** Springer Berlin Heidelberg pp. 2757–2760 https://doi.org/10.1007/978-3-642-16712-6_502 | Google Scholar

[63] Afek Ariel, Schipper Joshua L., Horton John, Gordân Raluca, Lukatsky David B. (2014) **Protein-dna binding in the absence of specific base-pair recognition** *Proceedings of the National Academy of Sciences* **111**:17140–17145 https://doi.org/10.1073/pnas.1410569111 | Google Scholar

Author information

Wencheng Ji

Department of Physics of Complex Systems, Weizmann Institute of Science, Rehovot, Israel, John A. Paulson School of Engineering and Applied Sciences, Harvard University, Cambridge, United States

For correspondence: wencheng.ji@weizmann.ac.il

Ori Hachmo

Department of Physics of Complex Systems, Weizmann Institute of Science, Rehovot, Israel

Naama Barkai

Department of Molecular Genetics, Weizmann Institute of Science, Rehovot, Israel

Ariel Amir

Department of Physics of Complex Systems, Weizmann Institute of Science, Rehovot, Israel, John A. Paulson School of Engineering and Applied Sciences, Harvard University, Cambridge, United States

For correspondence: ariel.amir@weizmann.ac.il

Editors

Reviewing Editor

Pierre Sens

Institut Curie, CNRS UMR168, Paris, France

Senior Editor

Aleksandra Walczak

CNRS, Paris, France

Reviewer #1 (Public review):

Summary:

The authors define the principles that, based on first principles, should be guiding the optimisation of transcription factors with intrinsically disordered regions (IDR). The authors introduce an original search process, coined "octopusing", that involves transcription factor IDR and their binding affinities to optimise search times and binding affinities. The first part concerns the optimal strategies to define binding affinities to the genome in the receiving region that is called the "antenna", highlighting the following: (i) reduce the target to IDR-binding distance on the genome, (ii) optimise the distance between the DNA binding domain and the binding sites on the IDR to be as close as possible to the distance between their binding sites on the genome; (iii) keep the same number of binding sites and their targets and modulate this number with binding strength, reducing them with increased strength; (iv) modulate the binding strength to be above a threshold that depends on the proportion of IDR binding sites in the antenna. The second part concerns the scaling of the search time in function of key parameters such as the volume of the nucleus, and the size of the antenna, derived as a combination of 3D search and 1D "octopusing". The third part focuses on validation, where the current results are compared to binding probability data from a single experiment, and new experiments are proposed to further validate the model as well as testing designed transcription factors.

Strengths:

The strength of this work is that it provides simple, interpretable and testable theoretical conclusions. This will allow the derived design principles to be understood, evaluated and improved in the future. The theoretical derivations are rigorous. The authors provide a comparison to experiments, and also propose new experiments to be performed in the future. This is a great value in the paper since it will set the stage and inspire new experimental techniques. Further, the field needs inspiration and motivation to develop these techniques, since they are required to benchmark the transcription factors designed with the methods presented in this paper, as well as to develop novel data based or in vivo methods that would greatly benefit the field. As such, this paper is a fundamental contribution to the field.

Weaknesses:

The model presents many first principles to drive the design of transcription factors, but arguably, other principles and mechanisms might also play a role by being beneficial to the search and binding process. These other principles are mentioned at the end of the discussion part of the paper. On the other hand, an important task left to do, is to critically consider these principles altogether, and analyse the available data to quantify which role is predominant among transcription factors IDRs functions. Further, since one function doesn't exclude another, a theoretical investigation of possible crosstalk, interaction, and cooperativity of those different hypothetical functions is still missing.

https://doi.org/10.7554/eLife.104956.2.sa2

Reviewer #2 (Public review):

Summary:

This is an interesting theoretical exploration of how a flexible protein domain, which has multiple DNA-binding sites along it, affects the stability of the protein-DNA complex. It proposes a mechanism ("octopusing") for protein doing a random walk while bound to DNA which simultaneously enables exploration of the DNA strand and stability of the bound state.

Strengths:

Stability of the protein-DNA bound state and the ability of the protein to perform 1d diffusion along the DNA are two properties of a transcription factor that are usually seen as being in opposition of each other. The octopusing mechanism is an elegant resolution of the puzzle of how both could be accommodated. This mechanism has interesting biological implications for the functional role of intrinsically disordered domains in transcription factor (TF) proteins. They show theoretically how these domains, if flexible and able to make multiple weak contacts with the DNA, can enhance the ability of the TF to efficiently find their binding site on the DNA from which they exert control over the transcription of their target gene. The paper concludes with a comparison of model predictions with experimental data which gives further support to the proposed mechanism. Overall, this is an interesting and well-executed theoretical paper that proposes an interesting idea about the functional role for IDR domains in TFs.

Weaknesses:

It is not clear how ubiquitous among eukaryotic transcription factors are the DNA binding sites for multiple subdomains along the IDR, which are assumed by the model. These assumptions though, provide interesting points of departure for further experiments.

https://doi.org/10.7554/eLife.104956.2.sa1

Author response:

The following is the authors' response to the original reviews

> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *The authors define the principles that, based on first principles, should be guiding the optimisation of trascription factors with intrinsically disordered regions (IDR). The first part of the study defines the following principles to optimize the binding affinities to the genome in the receiving region that is called the "antenna": (i) reduce the target to IDR-binding distance on the genome, (ii) optimise the distance betwee the DNA binding domain and the binding sites on the IDR to be as close as possible to the distance between their binding sites on the genome; (iii) keep the same number of binding sites and their targets and modulate this number with binding strength, reducing them with increased strenght; (iv) modulate the binding strenght to be above a threshold that depends on the proportion of IDR binding sites in the antenna. The second part defines the scaling of the seach time in function of key parameters such as the volume of the nucleus, and the size of the antenna, derived as a combination of 3D search of the antenna and 1D "octopusing" on the antenna. The third part focuses on validation, where the current results are compared to binding probabilith data from a single experiment, and new experiment are proposed to further validate the model as well as testing designed transcription factors.*
>
> *Strengths:*
>
> *The strength of this work is that it provides simple, interpretable and testable theoretical conclusions. This will allow the derived design principles to be understood, evaluated and improved in the future. The theoretical derivations are rigorous. The authors provides a comparison to experiments, and also propose new experiments to be performed in the future, this is a great value in the paper since it will set the stage and inspire new experimental techniques. Further, the field needs inspiration and motivations to develop these techniques, since they are required to benchmark the transcription factors designed with the methods presented in this paper, as well as to develop novel data based or in vivo methods that would greatly benefit the field. As such, this paper is a fundamental contribution to the field.*
>
> *Weaknesses:*
>
> *The model assumption that the interaction between the transcription factor and the DNA outside of the antenna region is negligible is probably too strong for many/most transcription factors, particularly in organisms with a longer genome than yeasts. The model presents many first principles to drive the design of transcription factor, but arguably, other principles and mechanisms might also play a role by being beneficial to*

the search and binding process. Specifically: (i) a role of the IDR in complex formation and cooperativity between multiple trascription factors, (ii) ability of the IDR to do parallel searching based on multiple DNA binding sites spaced by disordered regions, (iii) affinity of the IDR to specific compartmentalisations in the nucleus reducing the search time, etc. The paper would be improved by a discussion over alternative mechanisms.

We thank the reviewer for highlighting that our work delivers simple, interpretable and rigorously derived conclusions, backed by experimental comparison and concrete proposals for future studies.

Regarding interactions outside the antenna region, Supplementary S10 shows that the non-specific IDR–DNA interactions (on the order of 1 kBT) only slightly alter the 3D diffusion coefficient and thus do not affect our conclusions regarding the optimal search process.

We have also added sentences in the discussion section regarding the alternative mechanism.

Reviewer #2 (Public review):

Summary:

This is an interesting theoretical exploration of how a flexible protein domain, which has multiple DNAbinding sites along it, affects the stability of the protein-DNA complex. It proposes a mechanism ("octopusing") for protein doing a random walk while bound to DNA which simultaneously enables exploration of the DNA strand and stability of the bound state.

Strengths:

Stability of the protein-DNA bound state and the ability of the protein to perform 1d diffusion along the DNA are two properties of a transcription factor that are usually seen as being in opposition of each other. The octopusing mechanism is an elegant resolution of the puzzle of how both could be accommodated. This mechanism has interesting biological implications for the functional role of intrinsically disordered domains in transcription factor (TF) proteins. They show theoretically how these domains, if flexible and able to make multiple weak contacts with the DNA, can enhance the ability of the TF to efficiently find their binding site on the DNA from which they exert control over the transcription of their target gene. The paper concludes with a comparison of model predictions with experimental data which gives further support to the proposed model. Overall, this is an interesting and well executed theoretical paper that proposes an interesting idea about the functional role for IDR domains in TFs.

Weaknesses:

IDR domains are assumed flexible which I believe is not always the case. Also, I'm not sure how ubiquitous are the assumed binding sites on the DNA for multiple subdomains along the IDR. These assumptions though seem like interesting points of departure for further experiments.

We thank the reviewer for their careful and insightful evaluation of our work. In particular, we appreciate your emphasis on the inherent trade-off between binding stability and one-dimensional diffusion, and your recognition of how the octopusing mechanism elegantly reconciles these conflicting requirements.

To address the flexibility of TFs with IDRs, we incorporated the spring's rest length—effectively introducing tunable rigidity—in Supplementary Section S1, and we show that our design principles for binding probability remain robust. Indeed, this is a highly interesting point; a comprehensive study will require more detailed modeling alongside experimental validation.

We acknowledge that the current evidence for IDR-directed DNA binding is primarily derived from a limited number of well-studied cases, particularly Msn2 in yeast, and the ubiquity of this mechanism across diverse transcription factors remains to be established.

> ***Reviewer #1 (Recommendations for the authors):***
>
> *The paper jumps to fast to the results, an larger introduction might improve the paper, the current introduction jumps too fast to results. Further, line 50, I don't think that the figure is properly referenced. The formula 2 is confusing since what is the target volume V1 is not explained in the context of the formula, please expand the explanations.*

We appreciate the reviewer's valuable recommendations. We have expanded the Introduction, clarified V_1, and updated the line 50.

> ***Reviewer #2 (Recommendations for the authors):***
>
> *I have some mostly minor suggestions to the authors for improving the manuscript:*
>
> *In the abstract and introduction on at least two occasions the authors talk about IDRs as though they're necessarily flexible. My understanding is that, while this is a very reasonable assumption, I don't think this is something we know with any certainty for most IDRs. If the authors agree with my assessment I think they should reflect this uncertainty in the writing.*

Thank you for the recommendations. We revised the wording to reflect the uncertainty, changing it to: "... commonly assumed to behave as a long, flexible..." and "...can be assumed as flexible....".

> *It took me a bit of time to figure out what's going on in Figure 1b. To help the reader I would suggest labeling the DBD targets (yellow square) and the IDR targets (gray squares) as such. The figure also left me guessing whether the DBD domain can bind to the IDR targets non-specifically? (I presume not.) This also brought a slightly bigger question into focus for me, wouldn't the presence of the IDR binding "sites" (since these "sites" are on the protein I think the term "domains" instead of "sites") mean that this would increase the time the protein is bound non-specifically somewhere far from the target thereby increasing the search time. Or is the ability of the protein to bind specifically to DNA away from the DBD target ignored?*

We have labeled the DBD targets and IDR targets in the figure. 'Domains' usually refers to structured parts; we keep using 'sites' and clarify that they correspond to short linear motifs.

The reviewer is correct. Our model omits any non-specific binding between the DBD and IDR-binding targets, as well as between the TF and other DNA regions. If such interactions were to substantially lengthen the search time, they would effectively revert our mechanism to the classical bacterial facilitateddiffusion model, which is generally considered inappropriate for IDR-mediated TF search in eukaryotic cells. However, Supplementary Figure S10 demonstrates that non-specific IDR–DNA interactions induce only marginal changes in the effective three-dimensional diffusion coefficient within complex chromatin environments, and therefore do not alter our conclusions regarding the optimal search process.

> *In Equation 2 and the text that follows I was left wondering what is the target volume V1. Also, I think it would be helpful to the reader to give them a sense of scale for the dimension full quantities appearing in Equation 2. This is done later when comparing the*

> *theory to experimental data, but I think it would be helpful to give a sense of size earlier in the manuscript.*

V_1 denotes the volume of the IDR–binding target region, which is on the order of bp^3. $f(d,l_0)$ has units of inverse volume. We have included the units and specified the order of magnitude of V_1 after Equation 2.

> *The binding energy EB is discussed a number of times but it wasn't clear to me that this quantity referred to the energy per IDR site on the DNA or the total energy when the IDR is bound to DNA. In Figure 1 it would seem that the model allows only one IDR domain bound at a given time but I think the model allows for multiple IDR domains to be bound to the IDR target sites simultaneously. Right? Maybe make this clear in the Figure and the text.*

E_{B} denotes the binding energy per binding site, where each site corresponds to a short linear motif. Yes, we allow for multiple IDR domains to be bound to the IDR target sites simultaneously. We have clarified the definition of E_{B} and adjusted the figure slightly to avoid any misunderstanding.

> *After Eq 4 the discussion suggests that for Φ << 1 the threshold energy is much greater than kBT, but that's hard to imagine given that the logarithmic dependence of the latter on the former. Also in Figure 2d it seems that the threshold energy is about 8 kBT. Clearly this is not a big deal, just thought the authors might want to revise the language.*

Thank you. We now clarify the sentence using the representative values of Φ and E_{th} after Equation 4.

> *Right after Figure 2 there is a discussion of the different parameters that the authors vary. I suggest having a figure that illustrates these parameters (possibly in Figure 1b) to make it easier to follow the discussion.*

We have added explanations of the relevant parameters in Figure 1 for clarity.

> *When discussing the dynamics of search the result stated is that the search time is minimum for a specific value of R. I think it would be useful to translate this into a TF concentration. Also, if R represents the radius of the cells nucleus 1/6 um is almost an order of magnitude smaller than the size of a typical nucleus. Is this a worry? Either way some clarification of this number would be helpful.*

Thank you for the suggestion. As noted later in this section, we have translated R into an equivalent TF concentration, and we clarify that we assume the scaling of the minimum search time remains unchanged when extrapolated to the size of a typical nucleus.

> *There is a comment regarding the role of the DNA persistence length and how it was not accounted for. It would be helpful if the authors could add a sentence or two explains how a folded DNA conformation, as is the case in the nucleus, would affect their calculation. (So that the reader gets an idea without having to get into the details described in the Supplement).*

Thank you. We have revised the sentence to: "We have verified that reducing the DNA persistence length, which promotes increased DNA coiling, results in only a modest increase in mean search time. Even under extreme coiling conditions, the increase remains below 30% of the baseline value, as detailed in Supplementary S9.".

https://doi.org/10.7554/eLife.104956.2.sa0